

1 Metric spaces

1.1 Definitions and examples

Definition 1.1. A metric space is a pair (X, d) , where X is a set and $d : X \times X \rightarrow [0, \infty)$ is a function, called a metric, satisfying the following properties for all $x, y, z \in X$:

- (a) $d(x, y) = d(y, x)$;
 - (b) $d(x, y) = 0$ if and only if $x = y$;
 - (c) $d(x, y) \leq d(x, z) + d(z, y)$ (the triangle inequality).
- For $x \in X$ and $r > 0$, write

$$B(x; r) = \{y \in X : d(x, y) < r\}, \quad \overline{B}(x; r) = \{y \in X : d(x, y) \leq r\}.$$

The set $B(x; r)$ is the open ball centered at x with radius r , and $\overline{B}(x; r)$ is the closed ball centered at x with radius r .

Definition 1.2. Let (X, d) be a metric space. A subset $G \subseteq X$ is open if, for every $x \in G$, there is an $r > 0$ such that $B(x; r) \subseteq G$. A subset $F \subseteq X$ is closed if its complement $X \setminus F$ is open.

Definition 1.3. If (X, d) is a metric space and $Y \subseteq X$, then the restriction of d to $Y \times Y$ is a metric on Y . A subset $A \subseteq Y$ is relatively open (respectively, relatively closed) in Y if it is open (respectively, closed) in the metric space Y .

Theorem 1.4. Let (X, d) be a metric space and let $Y \subseteq X$.

(a) A subset $G \subseteq Y$ is relatively open in Y if and only if there is an open subset $U \subseteq X$ such that $G = U \cap Y$.

(b) A subset $F \subseteq Y$ is relatively closed in Y if and only if there is a closed subset $D \subseteq X$ such that $F = D \cap Y$.

Proof. (a) Suppose first that G is relatively open in Y . For each $x \in G$, choose $r_x > 0$ such that $B(x; r_x) \cap Y \subseteq G$, and set

$$U = \bigcup_{x \in G} B(x; r_x).$$

Then U is open and $U \cap Y = G$. Conversely, if $G = U \cap Y$ with U open in X , then for each $x \in G$ there is an $r > 0$ such that $B(x; r) \subseteq U$. Hence $B(x; r) \cap Y \subseteq G$, so G is relatively open in Y .

(b) The set F is relatively closed in Y if and only if $Y \setminus F$ is relatively open in Y . By part (a), this is equivalent to $Y \setminus F = U \cap Y$ for some open set $U \subseteq X$. In that case, with $D = X \setminus U$, we have

$$D \cap Y = (X \setminus U) \cap Y = Y \setminus (U \cap Y) = F.$$

The converse follows by reversing this argument. □

Theorem 1.5. Let (X, d) be a metric space.

(a) For all $x, y, z \in X$,

$$|d(x, y) - d(y, z)| \leq d(x, z).$$

- (b) A finite intersection of open sets is open.
- (c) An arbitrary union of open sets is open.
- (d) A finite union of closed sets is closed.
- (e) An arbitrary intersection of closed sets is closed.

Proof. For (a), the triangle inequality gives $d(x, y) \leq d(x, z) + d(z, y)$ and $d(y, z) \leq d(y, x) + d(x, z)$. Rearranging both inequalities gives the result. Parts (b)–(e) follow directly from the definitions and De Morgan's laws. $\square$

Definition 1.6. Let $A \subseteq X$. The interior of A is

$$\text{int } A = \bigcup \{G : G \text{ is open and } G \subseteq A\}.$$

The closure of A is

$$\text{cl } A = \bigcap \{F : F \text{ is closed and } A \subseteq F\}.$$

The boundary of A is

$$\partial A = \text{cl } A \cap \text{cl}(X \setminus A).$$

Theorem 1.7. Let $A \subseteq X$.

- (a) A point x belongs to $\text{int } A$ if and only if there is an $r > 0$ such that $B(x; r) \subseteq A$.
- (b) A point x belongs to $\text{cl } A$ if and only if $B(x; r) \cap A \neq \emptyset$ for every $r > 0$.

Proof. Part (a) follows immediately from the definition of an open set and the fact that $\text{int } A$ is the union of all open subsets of A .

For (b), if $B(x; r) \cap A = \emptyset$ for some $r > 0$, then $A \subseteq X \setminus B(x; r)$, where $X \setminus B(x; r)$ is closed and does not contain x . Thus $x \notin \text{cl } A$. Conversely, if $x \notin \text{cl } A$, then the open set $X \setminus \text{cl } A$ contains x , so some ball centered at x is disjoint from A . $\square$

Theorem 1.8. Let $A \subseteq X$.

- (a) A is closed if and only if $A = \text{cl } A$.
- (b) A is open if and only if $A = \text{int } A$.
- (c)

$$\text{cl } A = X \setminus \text{int}(X \setminus A), \quad \text{int } A = X \setminus \text{cl}(X \setminus A),$$

and

$$\partial A = \text{cl } A \setminus \text{int } A.$$

- (d) If $A_1, \dots, A_n \subseteq X$, then

$$\text{cl} \left(\bigcup_{k=1}^n A_k \right) = \bigcup_{k=1}^n \text{cl } A_k.$$

- (e) If $A_1, \dots, A_n \subseteq X$, then

$$\text{int} \left(\bigcap_{k=1}^n A_k \right) = \bigcap_{k=1}^n \text{int } A_k.$$

Proof. Parts (a) and (b) follow from the fact that $\text{cl } A$ is the smallest closed set containing A , whereas $\text{int } A$ is the largest open set contained in A .

For the first identity in (c), a point x is outside $\text{cl } A$ if and only if some open neighborhood of x is contained in $X \setminus A$; this is equivalent to $x \in \text{int}(X \setminus A)$. The second identity follows by replacing A by $X \setminus A$. Finally,

$$\partial A = \text{cl } A \cap \text{cl}(X \setminus A) = \text{cl } A \cap (X \setminus \text{int } A) = \text{cl } A \setminus \text{int } A.$$

For (e), the inclusion from left to right is immediate. Conversely, if $x \in \bigcap_{k=1}^n \text{int } A_k$, choose $r_k > 0$ such that $B(x; r_k) \subseteq A_k$ and set $r = \min\{r_1, \dots, r_n\}$. Then $B(x; r) \subseteq \bigcap_{k=1}^n A_k$. Part (d) follows from (c), (e), and De Morgan's laws. $\square$

Definition 1.9. If (X_1, d_1) and (X_2, d_2) are metric spaces, define a metric on $X_1 \times X_2$ by

$$d((x_1, x_2), (y_1, y_2)) = d_1(x_1, y_1) + d_2(x_2, y_2).$$

Theorem 1.10. *Adopt the preceding notation.*

- (a) *If $G_1 \subseteq X_1$ and $G_2 \subseteq X_2$ are open, then $G_1 \times G_2$ is open in $X_1 \times X_2$.*
- (b) *If $F_1 \subseteq X_1$ and $F_2 \subseteq X_2$ are closed, then $F_1 \times F_2$ is closed in $X_1 \times X_2$.*
- (c) *If G is open in $X_1 \times X_2$ and $(x_1, x_2) \in G$, then there is an $r > 0$ such that*

$$B_{d_1}(x_1; r) \times B_{d_2}(x_2; r) \subseteq G.$$

Proof. (a) If $(x_1, x_2) \in G_1 \times G_2$, choose $\varepsilon_i > 0$ such that $B_{d_i}(x_i; \varepsilon_i) \subseteq G_i$ for $i = 1, 2$. Then the ball in the product metric of radius $\min\{\varepsilon_1, \varepsilon_2\}$ is contained in $G_1 \times G_2$.

(b) We have

$$(X_1 \times X_2) \setminus (F_1 \times F_2) = ((X_1 \setminus F_1) \times X_2) \cup (X_1 \times (X_2 \setminus F_2)),$$

which is open by part (a).

(c) Choose $\varepsilon > 0$ such that $B_d((x_1, x_2); \varepsilon) \subseteq G$, and put $r = \varepsilon/2$. If $d_1(x_1, y_1) < r$ and $d_2(x_2, y_2) < r$, then $d((x_1, x_2), (y_1, y_2)) < \varepsilon$, proving the claim. $\square$

Definition 1.11. A subset E of a metric space (X, d) is dense if $\text{cl } E = X$. The metric space is separable if it has a countable dense subset.

Theorem 1.12. *A set E is dense in (X, d) if and only if $B(x; r) \cap E \neq \emptyset$ for every $x \in X$ and every $r > 0$.*

Proof. This is the characterization of closure by open balls applied to the identity $\text{cl } E = X$. $\square$

Exercise 1.13. Let ℓ^∞ denote the set of all bounded sequences of real numbers; that is,

$$\ell^\infty = \left\{ (x_n) : \sup_n |x_n| < \infty \right\}.$$

If $x = (x_n)$ and $y = (y_n)$ belong to ℓ^∞ , define

$$d(x, y) = \sup_n |x_n - y_n|.$$

(a) Show that d is a metric on ℓ^∞ .

(b) If e_n is the sequence with a 1 in the n th coordinate and zeros elsewhere, show that $B(e_n; 1/2) \cap B(e_m; 1/2) = \emptyset$ whenever $n \neq m$.

(c) Is the set $\{e_n : n \geq 1\}$ closed?

Proof. (a) Symmetry and positive definiteness are immediate. For every coordinate,

$$|x_n - y_n| \leq |x_n - z_n| + |z_n - y_n|.$$

Taking suprema gives $d(x, y) \leq d(x, z) + d(z, y)$.

(b) If x belonged to both balls and $n \neq m$, then $|x_n - 1| < 1/2$ because $d(x, e_n) < 1/2$, while $|x_n| < 1/2$ because the n th coordinate of e_m is zero. This is impossible.

(c) Yes. If a sequence of points from $A = \{e_n : n \geq 1\}$ converges, then it is Cauchy. Since $d(e_n, e_m) = 1$ whenever $n \neq m$, that sequence must be eventually constant. Its limit therefore belongs to A , so A is closed. $\square$

1.2 Sequences and completeness

Definition 1.14. A sequence (x_n) in X converges to x if, for every $\varepsilon > 0$, there is an integer N such that $d(x_n, x) < \varepsilon$ whenever $n \geq N$. We write $x_n \rightarrow x$.

If $A \subseteq X$, a point $x \in X$ is a limit point of A if every ball centered at x contains a point of A different from x .

For a set $A \subseteq X$, the distance from x to A is

$$\text{dist}(x, A) = \inf\{d(x, a) : a \in A\},$$

where we use the convention $\inf \emptyset = +\infty$. When A is nonempty, it is possible that $\text{dist}(x, A) = 0$ even though $x \notin A$.

A sequence (x_n) is Cauchy if, for every $\varepsilon > 0$, there is an integer N such that $d(x_n, x_m) < \varepsilon$ whenever $m, n \geq N$. A metric space is complete if every Cauchy sequence converges.

Theorem 1.15. Let (X, d) be a metric space.

(1) A subset $F \subseteq X$ is closed if and only if every convergent sequence of points of F has its limit in F .

(2) Every subsequence of a sequence converging to x also converges to x .

(3) Let $A \subseteq X$.

(a) A point x is a limit point of A if and only if there is a sequence of distinct points of A converging to x .

(b) The set A is closed if and only if it contains all its limit points.

(c)

$$\text{cl } A = A \cup \{x : x \text{ is a limit point of } A\}.$$

(4)

$$\text{cl } A = \{x \in X : \text{dist}(x, A) = 0\}.$$

(5) If (x_n) is Cauchy and a subsequence converges to x , then $x_n \rightarrow x$.

(6) The diameter of a nonempty set E is

$$\text{diam } E = \sup\{d(x, y) : x, y \in E\},$$

where the value may be $+\infty$. Moreover, $\text{diam } E = \text{diam}(\text{cl } E)$.

The proofs are the same as the standard proofs for subsets and sequences in $\mathbb{R}$, with absolute value replaced by the metric.

Theorem 1.16 (Cantor's Intersection Theorem). A metric space (X, d) is complete if and only if every sequence (F_n) of nonempty subsets satisfying

(i) each F_n is closed;

(ii) $F_1 \supseteq F_2 \supseteq \dots$;

(iii) $\text{diam } F_n \rightarrow 0$

has an intersection $\bigcap_{n=1}^{\infty} F_n$ consisting of exactly one point.

Proof. Suppose first that X is complete. Choose $x_n \in F_n$. If $m, n \geq N$, then $x_m, x_n \in F_N$, so $d(x_m, x_n) \leq \text{diam } F_N$. Thus (x_n) is Cauchy and converges to some $x \in X$. For each fixed N , all x_n with $n \geq N$ lie in the closed set F_N , so $x \in F_N$. Hence $x \in \bigcap_n F_n$. If x and y both belong to the intersection, then $d(x, y) \leq \text{diam } F_n$ for every n , so $x = y$.

Conversely, let (x_n) be Cauchy and set

$$F_n = \text{cl}\{x_m : m \geq n\}.$$

The sets F_n are nonempty, closed, and decreasing. The Cauchy property and $\text{diam}(\text{cl } E) = \text{diam } E$ imply $\text{diam } F_n \rightarrow 0$. Let $\{x\} = \bigcap_n F_n$. Since $x_n, x \in F_n$, we have $d(x_n, x) \leq \text{diam } F_n \rightarrow 0$. Thus $x_n \rightarrow x$, and X is complete. $\square$

Theorem 1.17. *Let (X, d) be complete and let $Y \subseteq X$. Then Y , with the restricted metric, is complete if and only if Y is closed in X .*

Proof. If Y is closed, every Cauchy sequence in Y converges in X , and its limit belongs to Y . Thus Y is complete.

Conversely, suppose Y is complete and (y_n) is a sequence in Y that converges in X to y . The sequence is Cauchy in Y , so it converges in Y to some $z \in Y$. Uniqueness of limits in X gives $y = z \in Y$. Therefore Y is closed. $\square$

Definition 1.18. A subset A of a metric space is bounded if $A = \emptyset$ or there are $x_0 \in X$ and $R > 0$ such that $A \subseteq B(x_0; R)$.

Theorem 1.19. (a) *A subset A of (X, d) is bounded if and only if, for every $x \in X$, there is an $r > 0$ such that $A \subseteq B(x; r)$.*

(b) *A finite union of bounded sets is bounded.*

(c) *The range of every Cauchy sequence is bounded.*

Proof. These statements follow directly from the triangle inequality. $\square$

Exercise 1.20. If $A \subseteq X$, show that

$$\text{int } A = \{x \in X : \text{dist}(x, X \setminus A) > 0\}.$$

Give an analogous characterization of ∂A .

Proof. A point x is in $\text{int } A$ if and only if some ball centered at x is contained in A , which is equivalent to $\text{dist}(x, X \setminus A) > 0$. Also,

$$\partial A = \{x \in X : \text{dist}(x, A) = 0 \text{ and } \text{dist}(x, X \setminus A) = 0\}.$$

$\square$

Exercise 1.21. Let (X, d) be the Cartesian product of (X_1, d_1) and (X_2, d_2) with the sum metric.

(a) Show that $((x_n^1, x_n^2))$ is Cauchy in X if and only if (x_n^1) is Cauchy in X_1 and (x_n^2) is Cauchy in X_2 .

(b) Show that X is complete if and only if both X_1 and X_2 are complete.

Proof. Part (a) follows from

$$d_i(x_n^i, x_m^i) \leq d((x_n^1, x_n^2), (x_m^1, x_m^2))$$

and from the definition of the sum metric. Part (b) follows from part (a) and the corresponding coordinatewise criterion for convergence. $\square$

1.3 Continuity

Definition 1.22. Let (X, d) and (Z, ρ) be metric spaces. A function $f : X \rightarrow Z$ is continuous at $a \in X$ if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$d(a, x) < \delta \implies \rho(f(a), f(x)) < \varepsilon.$$

The function is continuous on X if it is continuous at every point of X .

Theorem 1.23 (Sequential criterion). *Let (X, d) and (Z, ρ) be metric spaces and let $f : X \rightarrow Z$. Then f is continuous at a if and only if $x_n \rightarrow a$ implies $f(x_n) \rightarrow f(a)$ for every sequence (x_n) in X .*

Theorem 1.24. *Let $f : X \rightarrow Z$ be a function between metric spaces. The following are equivalent:*

- (a) f is continuous;
- (b) $f^{-1}(U)$ is open in X for every open set $U \subseteq Z$;
- (c) $f^{-1}(D)$ is closed in X for every closed set $D \subseteq Z$.

Proof. This is the usual inverse-image characterization of continuity; (b) and (c) are equivalent because inverse images commute with complements. $\square$

Theorem 1.25. *If A is a nonempty subset of the metric space (X, d) , then*

$$|\text{dist}(x, A) - \text{dist}(y, A)| \leq d(x, y)$$

for all $x, y \in X$.

Proof. For every $a \in A$,

$$d(x, a) \leq d(x, y) + d(y, a).$$

Taking the infimum over $a \in A$ gives

$$\text{dist}(x, A) \leq d(x, y) + \text{dist}(y, A).$$

Interchanging x and y yields the reverse inequality and hence the stated estimate. $\square$

Corollary 1.26. *For every nonempty $A \subseteq X$, the function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = \text{dist}(x, A)$ is continuous. In fact, it is 1-Lipschitz.*

Theorem 1.27. *If $f, g : X \rightarrow \mathbb{R}$ are continuous, then $f + g$ and fg are continuous. If $f(x) \neq 0$ for every $x \in X$, then $1/f$ is continuous.*

Theorem 1.28. *The composition of continuous functions is continuous.*

Theorem 1.29 (Urysohn's Lemma for metric spaces). *If A and B are disjoint closed subsets of a metric space X , then there is a continuous function $f : X \rightarrow [0, 1]$ such that $f = 0$ on A and $f = 1$ on B .*

Proof. If $A = \emptyset$, take $f \equiv 1$; if $B = \emptyset$, take $f \equiv 0$. Otherwise, define

$$f(x) = \frac{\text{dist}(x, A)}{\text{dist}(x, A) + \text{dist}(x, B)}.$$

The denominator is positive at every point: if both distances were zero, the point would belong to both closed sets. The preceding distance theorem and the algebra of continuous functions show that f is continuous. Its values on A and B are as required. $\square$

Corollary 1.30. *If F is closed and G is open with $F \subseteq G$, then there is a continuous function $f : X \rightarrow [0, 1]$ such that $f = 1$ on F and $f = 0$ on $X \setminus G$.*

Proof. Apply the preceding theorem to the disjoint closed sets $X \setminus G$ and F . □

Definition 1.31. A map $f : (X, d) \rightarrow (Z, \rho)$ is a homeomorphism if it is bijective and both f and f^{-1} are continuous. Two metric spaces are homeomorphic if there is a homeomorphism between them.

Two metrics d and ρ on the same set X are equivalent if they induce the same topology. For metric spaces, this is equivalent to saying that they have the same convergent sequences, or that the identity map $(X, d) \rightarrow (X, \rho)$ is a homeomorphism.

Theorem 1.32. *For every metric space (X, d) ,*

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

defines an equivalent metric on X .

Proof. Symmetry and positive definiteness are immediate. Let $\phi(t) = t/(1+t)$. Since ϕ is increasing and

$$\phi(a + b) \leq \phi(a) + \phi(b) \quad (a, b \geq 0),$$

the triangle inequality for d implies the triangle inequality for ρ . Thus ρ is a metric.

Moreover, $\rho(x, y) \leq d(x, y)$. Conversely, for every $\delta > 0$,

$$\rho(x, y) < \frac{\delta}{1 + \delta} \implies d(x, y) < \delta.$$

Hence the identity maps in both directions are continuous, so the metrics are equivalent. □

Remark. The metric ρ is bounded by 1. Thus boundedness is not a topological property.

Remark. Equivalent metrics preserve continuity but need not preserve Cauchy sequences or completeness.

Example 1.33. Let

$$X = \{(x, y) \in \mathbb{R}^2 : x^2 + (y - 1)^2 = 1\}$$

with its Euclidean subspace metric. For $t \in \mathbb{R}$, take the line through $(t, 0)$ and $(0, 2)$ and let $f(t)$ be its intersection with X other than $(0, 2)$. Then f is a homeomorphism from $\mathbb{R}$ onto $Y = X \setminus \{(0, 2)\}$. Define

$$\rho(s, t) = d(f(s), f(t)).$$

The metric ρ is equivalent to the usual metric on $\mathbb{R}$, and $f : (\mathbb{R}, \rho) \rightarrow (Y, d)$ is an isometry. The sequence (n) is Cauchy with respect to ρ , because $f(n) \rightarrow (0, 2)$ in X , but it has no limit in $(\mathbb{R}, \rho)$. Therefore $(\mathbb{R}, \rho)$ is not complete.

Definition 1.34. A function $f : (X, d) \rightarrow (Z, \rho)$ is uniformly continuous if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$d(x, y) < \delta \implies \rho(f(x), f(y)) < \varepsilon$$

for all $x, y \in X$.

Theorem 1.35. (a) *A uniformly continuous function sends Cauchy sequences to Cauchy sequences.*
(b) *If $A \subseteq X$, Z is complete, and $f : A \rightarrow Z$ is uniformly continuous, then f has a unique uniformly continuous extension $\bar{f} : \text{cl } A \rightarrow Z$.*

Proof. Part (a) follows directly from the definitions. For (b), if $x \in \text{cl } A$, choose a sequence (a_n) in A with $a_n \rightarrow x$ and define

$$\bar{f}(x) = \lim_{n \rightarrow \infty} f(a_n).$$

Part (a) and completeness of Z guarantee that the limit exists. Uniform continuity shows that the value is independent of the chosen sequence and that $\bar{f}$ is uniformly continuous. Density of A gives uniqueness. $\square$

Exercise 1.36. (a) If $f : X \rightarrow Z$ is continuous, A is dense in X , and $f(a) = z$ for every $a \in A$, show that $f(x) = z$ for every $x \in X$.

(b) If $f : X \rightarrow Z$ is continuous and surjective and A is dense in X , show that $f(A)$ is dense in Z .

Proof. (a) For any $x \in X$, choose a sequence (a_n) in A with $a_n \rightarrow x$. Then $f(x) = \lim f(a_n) = z$.
(b) Let $z \in Z$ and let V be an open neighborhood of z . Choose $x \in X$ with $f(x) = z$. Then $f^{-1}(V)$ is an open neighborhood of x , so it meets A . Hence V meets $f(A)$. $\square$

1.4 Compactness

Definition 1.37. A collection $\mathcal{G}$ of subsets of X is a cover of $E \subseteq X$ if $E \subseteq \bigcup \mathcal{G}$. It is an open cover if every member of $\mathcal{G}$ is open. A subcover is a subcollection that still covers E .

A subset K of a metric space is compact if every open cover of K has a finite subcover.

Theorem 1.38. *Let (X, d) be a metric space.*

- (a) *Every compact subset of X is closed and bounded.*
- (b) *Every closed subset of a compact set is compact.*
- (c) *The continuous image of a compact set is compact.*

Corollary 1.39. *If X is nonempty and compact and $f : X \rightarrow \mathbb{R}$ is continuous, then f attains a minimum and a maximum on X .*

Definition 1.40. A subset K of a metric space is totally bounded if, for every $r > 0$, it can be covered by finitely many balls of radius r .

A collection $\mathcal{F}$ of sets has the finite intersection property (FIP) if every finite subcollection has nonempty intersection.

Theorem 1.41. *For a subset K of a metric space, the following are equivalent:*

- (a) *K is compact;*
- (b) *every collection of closed subsets of K having the FIP has nonempty intersection;*
- (c) *every sequence in K has a convergent subsequence whose limit belongs to K ;*
- (d) *every infinite subset of K has a limit point in K ;*
- (e) *K , with its subspace metric, is complete and totally bounded.*

Proof. (a) $\Rightarrow$ (b): If a family $\mathcal{F}$ of closed subsets of K has empty intersection, then the relative-open sets $K \setminus F$, $F \in \mathcal{F}$, cover K . A finite subcover would contradict the FIP.

(a) $\Rightarrow$ (d): If an infinite set $E \subseteq K$ had no limit point in K , every point of K would have a neighborhood meeting E in at most one point. A finite subcover of these neighborhoods would force E to be finite.

(d) $\Rightarrow$ (c): If a sequence has finite range, it has a constant subsequence. If its range is infinite, let x be a limit point of the range. Every neighborhood of x contains infinitely many terms of the sequence, so one can choose a subsequence converging to x .

(c) $\Rightarrow$ (e): A Cauchy sequence has a convergent subsequence; the whole sequence then converges to the same limit, so K is complete. If K were not totally bounded, there would be an $\varepsilon > 0$ and a sequence in K whose distinct terms are at least ε apart. Such a sequence has no convergent subsequence, a contradiction.

(e) $\Rightarrow$ (c): Repeatedly cover K by finitely many balls of radii $1, 1/2, 1/3, \dots$ and choose nested subsequences lying in one ball at each stage. A diagonal subsequence is Cauchy and therefore converges in K .

Finally, (c) $\Rightarrow$ (a). First note the Lebesgue-number property: for every open cover $\mathcal{G}$ of K , there is an $r > 0$ such that every ball $B(x; r)$ with $x \in K$ is contained in some member of $\mathcal{G}$. Otherwise, choose $x_n \in K$ such that $B(x_n; 1/n)$ is contained in no member of the cover. A convergent subsequence tends to some $x \in K$, and an open member of the cover containing x eventually contains the corresponding balls, a contradiction. Since (c) implies total boundedness by the preceding argument, finitely many balls of radius r cover K , and the corresponding members of $\mathcal{G}$ form a finite subcover. $\square$

Remark. Compact sets are totally bounded, but total boundedness alone does not imply compactness; completeness is also required.

Theorem 1.42 (Heine–Borel). *A subset of $\mathbb{R}^q$ is compact if and only if it is closed and bounded.*

Theorem 1.43. *Every continuous map from a compact metric space into a metric space is uniformly continuous.*

Theorem 1.44. *Every compact metric space is separable.*

Proof. For each $n \geq 1$, compactness gives a finite set $F_n \subseteq X$ such that

$$X = \bigcup_{x \in F_n} B(x; 1/n).$$

The set $F = \bigcup_{n=1}^{\infty} F_n$ is countable. Given $x \in X$ and $\varepsilon > 0$, choose n with $1/n < \varepsilon$ and then choose $y \in F_n$ with $d(x, y) < 1/n$. Thus F is dense. $\square$

Exercise 1.45. Show that the closure of a totally bounded set is totally bounded.

Proof. Let E be totally bounded and let $r > 0$. Choose $x_1, \dots, x_n \in E$ such that

$$E \subseteq \bigcup_{i=1}^n B(x_i; r/2).$$

Then

$$\text{cl } E \subseteq \bigcup_{i=1}^n \overline{B}(x_i; r/2) \subseteq \bigcup_{i=1}^n B(x_i; r),$$

so $\text{cl } E$ is totally bounded. $\square$

Exercise 1.46. If (X, d) is complete and $E \subseteq X$, show that E is totally bounded if and only if $\text{cl } E$ is compact.

Proof. If E is totally bounded, then $\text{cl } E$ is totally bounded by the preceding exercise. It is closed in the complete space X , so it is complete; hence it is compact. Conversely, if $\text{cl } E$ is compact, then it is totally bounded, and every subset of a totally bounded set is totally bounded. $\square$

1.5 Connectedness

Definition 1.47. A metric space X is connected if its only subsets that are both open and closed are X and $\emptyset$. Equivalently, X cannot be written as the union of two disjoint nonempty open sets. A subset of a metric space is connected when it is connected with its subspace metric.

Theorem 1.48. A subset of $\mathbb{R}$ is connected if and only if it is an interval.

Theorem 1.49. The continuous image of a connected set is connected.

Proof. If $f(X) = C \cup D$ were a separation of $f(X)$, then $f^{-1}(C)$ and $f^{-1}(D)$ would form a separation of X . $\square$

Corollary 1.50 (Intermediate Value Theorem). If X is connected, $f : X \rightarrow \mathbb{R}$ is continuous, and $a, b \in f(X)$ with $a < b$, then every $c \in [a, b]$ belongs to $f(X)$.

Theorem 1.51. Let X be a metric space.

(a) If $\{E_i : i \in I\}$ is a collection of connected subsets such that $E_i \cap E_j \neq \emptyset$ for all $i, j \in I$, then $\bigcup_{i \in I} E_i$ is connected.

(b) If (E_n) is a sequence of connected subsets such that $E_n \cap E_{n+1} \neq \emptyset$ for every n , then $\bigcup_{n=1}^{\infty} E_n$ is connected.

Proof. For (a), let A be a nonempty relatively clopen subset of $E = \bigcup_i E_i$. It meets some E_{i_0} , so connectedness gives $E_{i_0} \subseteq A$. Every E_j meets E_{i_0} , hence meets A , and therefore $E_j \subseteq A$. Thus $A = E$.

For (b), the finite union $F_n = E_1 \cup \cdots \cup E_n$ is connected by induction. Since all the F_n contain E_1 , part (a) shows that $\bigcup_n F_n = \bigcup_n E_n$ is connected. $\square$

Corollary 1.52. The union of two intersecting connected subsets is connected.

Definition 1.53. A component of a metric space is a maximal connected subset.

Remark. Maximal means maximal with respect to inclusion.

Theorem 1.54. Every connected subset of a metric space is contained in a component; distinct components are disjoint; and the union of all components is the whole space.

Proof. Let D be connected. Take the union C of all connected subsets that contain D . Every two such sets meet in D , so C is connected. It is maximal by construction and hence is a component containing D . Applying this to the singleton $\{x\}$ shows that every point lies in a component. If two components meet, their union is connected, so maximality forces them to be equal. $\square$

Theorem 1.55. If C is connected and $C \subseteq Y \subseteq \text{cl } C$, then Y is connected.

Proof. If $Y = A \cup B$ were a separation, then connectedness of C would imply that C is contained entirely in one of the two sets, say A . But C is dense in Y , so the nonempty relatively open set B would have to meet C , a contradiction. $\square$

Corollary 1.56. The closure of a connected set is connected, and every component is closed.

Definition 1.57. Let $E \subseteq X$, let $x, y \in E$, and let $\varepsilon > 0$. An ε -chain from x to y in E is a finite sequence $x = x_1, \dots, x_n = y$ in E such that

- (i) $B(x_k; \varepsilon) \subseteq E$ for every k ;
- (ii) $d(x_{k-1}, x_k) < \varepsilon$ for $2 \leq k \leq n$.

Theorem 1.58. *Let G be an open subset of $\mathbb{R}^q$.*

(a) *Every component of G is open, and G has at most countably many components.*

(b) *The set G is connected if and only if, for every $x, y \in G$, there is an $\varepsilon > 0$ and an ε -chain in G from x to y .*

Proof. (a) Let H be a component of G and $x \in H$. Since G is open, some ball $B(x; r)$ lies in G . The ball is connected and meets H , so $B(x; r) \subseteq H$. Thus H is open. Every nonempty open subset of $\mathbb{R}^q$ contains a point of $\mathbb{Q}^q$, and distinct components are disjoint; assigning a rational point to each component proves countability.

(b) Suppose first that G is connected. Fix $x \in G$ and let D be the set of points that can be joined to x by a polygonal path in G . Small balls show that both D and $G \setminus D$ are open in G , so $D = G$. Let γ be a polygonal path from x to y . Its compact image has positive distance from $\mathbb{R}^q \setminus G$. Uniform continuity of γ allows a sufficiently fine subdivision of the path whose vertices form an ε -chain for some $\varepsilon > 0$.

Conversely, the union of the balls appearing in an ε -chain is a connected subset of G : consecutive balls intersect because the distance between their centers is less than ε . Hence the endpoints of every chain lie in the same component. If such a chain exists between every pair of points, G has only one component and is connected. $\square$

1.6 The Baire Category Theorem

Theorem 1.59 (Baire Category Theorem). *If (X, d) is complete and (U_n) is a sequence of open dense subsets of X , then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .*

Proof. Let G be a nonempty open subset of X . Since U_1 is dense, $G \cap U_1$ is a nonempty open set. Choose $x_1 \in X$ and $r_1 < 1$ such that

$$\overline{B}(x_1; r_1) \subseteq G \cap U_1.$$

Inductively, after choosing x_{n-1} and r_{n-1} , the set $B(x_{n-1}; r_{n-1}) \cap U_n$ is nonempty and open. Choose x_n and $0 < r_n < 1/n$ such that

$$\overline{B}(x_n; r_n) \subseteq B(x_{n-1}; r_{n-1}) \cap U_n.$$

For $m > n$, the nesting gives $x_m \in B(x_n; r_n)$, so (x_n) is Cauchy. Let $x_n \rightarrow x$. For every fixed n , all sufficiently late terms lie in the closed set $\overline{B}(x_n; r_n)$, hence $x \in \overline{B}(x_n; r_n)$. Therefore

$$x \in G \cap \bigcap_{n=1}^{\infty} U_n.$$

Since every nonempty open G meets the intersection, the intersection is dense. $\square$

Corollary 1.60. *If a nonempty complete metric space is a countable union $X = \bigcup_{n=1}^{\infty} F_n$ of closed sets, then $\text{int } F_n \neq \emptyset$ for at least one n .*

Proof. If every F_n had empty interior, then each open set $U_n = X \setminus F_n$ would be dense. The Baire Category Theorem would imply that $\bigcap_n U_n$ is dense, but this intersection is empty. $\square$

Remark. The assumption that each U_n is open is essential.

Exercise 1.61. A set E is nowhere dense if $\text{int}(\text{cl } E) = \emptyset$. If $A = \bigcup_{n=1}^{\infty} E_n$ and every E_n is nowhere dense, show that $X \setminus A$ is dense in every complete metric space X .

Proof. For each n , the set

$$U_n = X \setminus \text{cl } E_n$$

is open and dense. Hence $\bigcap_n U_n$ is dense by the Baire Category Theorem. Since

$$\bigcap_n U_n = X \setminus \bigcup_n \text{cl } E_n \subseteq X \setminus \bigcup_n E_n = X \setminus A,$$

the set $X \setminus A$ is dense as well. □

2 Topological spaces

2.1 Definitions and examples

Definition 2.1. A topological space is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T}$ is a collection of subsets of X satisfying:

- (a) $\emptyset, X \in \mathcal{T}$;
- (b) if $\{G_i : i \in I\} \subseteq \mathcal{T}$, then $\bigcup_{i \in I} G_i \in \mathcal{T}$;
- (c) if $G_1, \dots, G_n \in \mathcal{T}$, then $\bigcap_{k=1}^n G_k \in \mathcal{T}$.

The collection $\mathcal{T}$ is called a topology on X , and its members are called open sets.

For any set X , the collection

$$\mathcal{T} = \{G \subseteq X : X \setminus G \text{ is finite}\} \cup \{\emptyset\}$$

is the *cofinite topology*. If X is infinite, this topology is not Hausdorff.

If $(X, \mathcal{T})$ is a topological space and $Y \subseteq X$, then

$$\mathcal{T}_Y = \{Y \cap G : G \in \mathcal{T}\}$$

is a topology on Y , called the subspace topology or relative topology. For a metric space, this agrees with the topology induced by the restricted metric.

A topological space X is Hausdorff if, whenever $x \neq y$, there are disjoint open sets U and V such that $x \in U$ and $y \in V$.

Agreement. Unless explicitly stated otherwise, all topological spaces below are assumed to be Hausdorff.

A subset F of a topological space X is closed if $X \setminus F$ is open.

Theorem 2.2. Let X be a topological space, and let $\mathcal{F}$ be the collection of closed subsets of X .

- (a) $\emptyset, X \in \mathcal{F}$.
- (b) An arbitrary intersection of members of $\mathcal{F}$ belongs to $\mathcal{F}$.
- (c) A finite union of members of $\mathcal{F}$ belongs to $\mathcal{F}$.
- (d) If X is Hausdorff, then every singleton $\{x\}$ is closed.

Proof. Parts (a)–(c) follow from De Morgan's laws. For (d), fix $x \in X$. For every $y \neq x$, choose an open set U_y containing y and disjoint from some open neighborhood of x . Then $X \setminus \{x\} = \bigcup_{y \neq x} U_y$ is open. □

Definition 2.3. Let $A \subseteq X$. Its interior, closure, and boundary are

$$\text{int } A = \bigcup \{G : G \text{ is open and } G \subseteq A\},$$

$$\text{cl } A = \bigcap \{F : F \text{ is closed and } A \subseteq F\},$$

and

$$\partial A = \text{cl } A \cap \text{cl}(X \setminus A).$$

Theorem 2.4. *Let $A \subseteq X$.*

- (a) $x \in \text{int } A$ if and only if there is an open set G such that $x \in G \subseteq A$.
- (b) $x \in \text{cl } A$ if and only if every open set G containing x meets A .
- (c) $\text{int } A$ is the largest open subset of A .
- (d) $\text{cl } A$ is the smallest closed set containing A .

Proof. All four statements follow directly from the definitions. □

Theorem 2.5. *Let $A \subseteq X$.*

- (a) A is closed if and only if $A = \text{cl } A$.
- (b) A is open if and only if $A = \text{int } A$.
- (c) If $A_1, \dots, A_n \subseteq X$, then

$$\text{cl} \left(\bigcup_{k=1}^n A_k \right) = \bigcup_{k=1}^n \text{cl } A_k.$$

Proof. Parts (a) and (b) follow from the extremal properties in the preceding theorem. For (c), the right-hand side is a closed set containing the union, which gives one inclusion. Conversely, each A_k is contained in the union, so each $\text{cl } A_k$ is contained in the closure of the union. □

Definition 2.6. A subset E of a topological space is dense if $\text{cl } E = X$. A topological space is separable if it has a countable dense subset.

Theorem 2.7. *A set E is dense in X if and only if every nonempty open subset of X meets E . Equivalently, every open neighborhood of every point meets E .*

Definition 2.8. A point $x \in X$ is a limit point of $A \subseteq X$ if every open neighborhood G of x satisfies

$$(G \setminus \{x\}) \cap A \neq \emptyset.$$

Theorem 2.9. *Let $A \subseteq X$.*

- (a) The set A is closed if and only if it contains all its limit points.
- (b)

$$\text{cl } A = A \cup \{x : x \text{ is a limit point of } A\}.$$

Proof. If $x \in \text{cl } A \setminus A$, then every neighborhood of x meets A , and any such point of A is different from x ; hence x is a limit point. Conversely, every limit point belongs to $\text{cl } A$ because every neighborhood of it meets A . This proves (b), and (a) follows. □

Theorem 2.10. *Let $Y \subseteq X$ have the subspace topology.*

- (a) A subset $A \subseteq Y$ is closed in Y if and only if $A = F \cap Y$ for some closed set $F \subseteq X$.
- (b) If $A \subseteq Y$, then

$$\text{cl}_Y A = Y \cap \text{cl}_X A.$$

- (c) If $A \subseteq Y$, then

$$Y \cap \text{int}_X A \subseteq \text{int}_Y A.$$

The inclusion can be strict.

Proof. (a) The set A is closed in Y if and only if $Y \setminus A = Y \cap G$ for some open $G \subseteq X$. This is equivalent to $A = Y \cap (X \setminus G)$.

(b) The set $Y \cap \text{cl}_X A$ is closed in Y and contains A . If C is any closed subset of Y containing A , write $C = Y \cap F$ with F closed in X . Then $\text{cl}_X A \subseteq F$, so $Y \cap \text{cl}_X A \subseteq C$.

(c) If $x \in Y \cap \text{int}_X A$, choose an open set $G \subseteq X$ with $x \in G \subseteq A$. Then $G \cap Y$ is an open neighborhood of x in Y contained in A . $\square$

2.2 Bases and subbases

Definition 2.11. A collection $\mathcal{B}$ of subsets of X is a base for a topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

Theorem 2.12. A collection $\mathcal{B}$ of subsets of X is a base for a topology on X if and only if:

- (a) $X = \bigcup_{B \in \mathcal{B}} B$;
 - (b) whenever $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, there is a $B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2$.
- The topology generated by $\mathcal{B}$ is Hausdorff if and only if, in addition,
- (c) for distinct $x, y \in X$, there are disjoint $A, B \in \mathcal{B}$ with $x \in A$ and $y \in B$.

Proof. If $\mathcal{B}$ is a base, (a) and (b) follow from the fact that X and $B_1 \cap B_2$ are open. Conversely, let $\mathcal{T}$ be the collection of all unions of members of $\mathcal{B}$. Conditions (a) and (b) give the empty set, the whole space, arbitrary unions, and finite intersections required of a topology. Condition (c) is exactly the Hausdorff separation condition expressed using basic neighborhoods. $\square$

Corollary 2.13. Let (X, d) be separable, let $\{a_1, a_2, \dots\}$ be dense in X , and let $\{r_1, r_2, \dots\}$ enumerate the positive rational numbers. Then

$$\mathcal{B} = \{B(a_n; r_m) : n, m \geq 1\}$$

is a countable base for the metric topology.

Proof. Let G be open and $x \in G$. Choose $\varepsilon > 0$ with $B(x; \varepsilon) \subseteq G$. Pick a_n with $d(x, a_n) < \varepsilon/3$, and choose a positive rational r_m satisfying

$$d(x, a_n) < r_m < \varepsilon - d(x, a_n).$$

Then $x \in B(a_n; r_m) \subseteq G$. $\square$

Corollary 2.14. Let $\mathcal{S}$ be any collection of subsets of X . The collection $\mathcal{B}$ of all finite intersections of members of $\mathcal{S}$, including the empty intersection X , is a base for a topology on X . This topology is Hausdorff if $\mathcal{S}$ separates points in the following sense: for every $x \neq y$, there are disjoint $S, T \in \mathcal{S}$ with $x \in S$ and $y \in T$.

Proof. Finite intersections cover X because the empty intersection is X , and the intersection of two finite intersections is again a finite intersection. The final assertion follows from part (c) of the basis theorem. $\square$

Definition 2.15. A collection $\mathcal{S}$ of subsets of X is called a subbase if the finite intersections of its members form a base. The resulting topology is the topology generated by $\mathcal{S}$.

Exercise 2.16. A partially ordered set is a pair $(X, \leq)$ in which $\leq$ is reflexive, transitive, and antisymmetric. It is linearly ordered if any two points are comparable.

For $a \in X$, put

$$(-\infty, a) = \{x \in X : x < a\}, \quad (a, \infty) = \{x \in X : a < x\}.$$

Let $\mathcal{S}$ be the collection of all such rays.

(a) Show that $\mathcal{S}$ generates a topology on every partially ordered set. This topology need not be Hausdorff.

(b) Show that it is Hausdorff when the order is linear.

(c) In the linearly ordered case, show that a base consists of the open intervals (a, b) together with the rays $(-\infty, b)$ and (a, ∞) that are needed at endpoints.

Proof. Part (a) follows from the subbase construction. For (b), let $x < y$. If there is a point z with $x < z < y$, then $(-\infty, z)$ and (z, ∞) separate x and y . If no such z exists, then $(-\infty, y)$ and (x, ∞) do. Part (c) follows by listing the possible nonempty finite intersections of left and right rays. $\square$

2.3 Continuous functions

Definition 2.17. Let X and W be topological spaces. A function $f : X \rightarrow W$ is continuous at $x \in X$ if, for every neighborhood U of $f(x)$, there is a neighborhood G of x such that $f(G) \subseteq U$. It is continuous if it is continuous at every point.

Theorem 2.18. For a function $f : X \rightarrow W$, the following are equivalent:

- (a) f is continuous;
- (b) $f^{-1}(U)$ is open in X for every open $U \subseteq W$;
- (c) $f^{-1}(C)$ is closed in X for every closed $C \subseteq W$;
- (d) if $\mathcal{B}$ is a base for W , then $f^{-1}(B)$ is open for every $B \in \mathcal{B}$;
- (e) if $\mathcal{S}$ is a subbase for W , then $f^{-1}(S)$ is open for every $S \in \mathcal{S}$.

Proof. The equivalence of (a) and (b) follows from the neighborhood definition. The equivalence of (b) and (c) follows because inverse images commute with complements. Clearly (b) implies (d), and (d) implies (e) because subbasic sets are open. Finally, if (e) holds, inverse images of finite intersections of subbasic sets are open, and inverse images commute with arbitrary unions; therefore the inverse image of every open set is open. $\square$

Theorem 2.19. The composition of continuous functions is continuous.

Proof. If $f : X \rightarrow Y$ and $g : Y \rightarrow W$ are continuous and $U \subseteq W$ is open, then

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$$

is open. $\square$

Theorem 2.20 (Pasting Lemma). Let $X = A \cup B$, and let $g : A \rightarrow W$ and $h : B \rightarrow W$ be continuous functions that agree on $A \cap B$. Define $f : X \rightarrow W$ by $f|_A = g$ and $f|_B = h$. If A and B are both open, or if they are both closed, then f is continuous.

Proof. If A and B are open and $U \subseteq W$ is open, then $g^{-1}(U)$ is open in A and hence in X ; similarly for $h^{-1}(U)$. Thus $f^{-1}(U) = g^{-1}(U) \cup h^{-1}(U)$ is open.

If A and B are closed, apply the same argument to inverse images of closed sets: a set closed in A is closed in X because A is closed, and similarly for B . $\square$

Theorem 2.21. Let X_k be a topological space for $1 \leq k \leq n$, and put $X = X_1 \times \cdots \times X_n$. The collection

$$\mathcal{B} = \{G_1 \times \cdots \times G_n : G_k \text{ is open in } X_k\}$$

is a base for a topology on X . This topology is the smallest topology for which every coordinate projection $\pi_k : X \rightarrow X_k$ is continuous.

Proof. The basis conditions are immediate. For an open set $U \subseteq X_k$,

$$\pi_k^{-1}(U) = X_1 \times \cdots \times X_{k-1} \times U \times X_{k+1} \times \cdots \times X_n$$

is basic open, so every projection is continuous. Conversely, if a topology on X makes all projections continuous, then

$$G_1 \times \cdots \times G_n = \bigcap_{k=1}^n \pi_k^{-1}(G_k)$$

is open. Thus it contains the product topology. □

Definition 2.22. The topology in the preceding theorem is the product topology. The maps π_k are the coordinate projections.

Definition 2.23. A function $f : X \rightarrow Z$ is an open map if $f(G)$ is open in Z whenever G is open in X .

Theorem 2.24. Every coordinate projection from a finite product with the product topology is an open map.

Proof. If the product is empty, the assertion is immediate. Otherwise, every factor is nonempty, and the image under π_k of a basic open set $G_1 \times \cdots \times G_n$ is either empty or G_k . Since every open set is a union of basic open sets and projections commute with unions, π_k is open. □

Theorem 2.25. Let Y be a topological space and let $f : Y \rightarrow X_1 \times \cdots \times X_n$. Then f is continuous if and only if each coordinate function $\pi_k \circ f$ is continuous.

Proof. One direction follows by composition. Conversely, for a basic open set $G_1 \times \cdots \times G_n$,

$$f^{-1}(G_1 \times \cdots \times G_n) = \bigcap_{k=1}^n (\pi_k \circ f)^{-1}(G_k),$$

which is open. □

Theorem 2.26. If $f, g : X \rightarrow \mathbb{R}$ are continuous, then $f + g$ and fg are continuous.

Proof. The map $x \mapsto (f(x), g(x))$ is continuous by the product criterion. Compose it with the continuous maps $(a, b) \mapsto a + b$ and $(a, b) \mapsto ab$. □

Theorem 2.27. If $f, g : X \rightarrow \mathbb{R}$ are continuous, then so are $f \vee g$, $f \wedge g$, and $|f|$.

Proof. The absolute-value function is continuous, and

$$f \vee g = \frac{f + g + |f - g|}{2}, \quad f \wedge g = \frac{f + g - |f - g|}{2}.$$

□

Definition 2.28. A homeomorphism between topological spaces X and Z is a bijection $f : X \rightarrow Z$ such that both f and f^{-1} are continuous.

2.4 Compactness and connectedness

Definition 2.29. A subset K of a topological space X is compact if every open cover of K has a finite subcover.

Theorem 2.30. Let X be Hausdorff and let $K \subseteq X$.

- (a) If K is compact, then K is closed.
- (b) If K is compact and $F \subseteq K$ is closed in K , then F is compact.
- (c) The continuous image of a compact set is compact.

Proof. (a) Fix $x \notin K$. For each $z \in K$, choose disjoint open sets U_z and V_z with $x \in U_z$ and $z \in V_z$. Finitely many $V_{z_1}, \dots, V_{z_n}$ cover K . Then $U = \bigcap_{k=1}^n U_{z_k}$ is an open neighborhood of x disjoint from K . Hence $X \setminus K$ is open.

(b) By part (a), K is closed in X . Since F is closed in K , it is closed in X . Add $X \setminus F$ to any open cover of F to obtain an open cover of K .

(c) Pull an open cover of $f(K)$ back under f , choose a finite subcover of K , and push it forward. $\square$

Corollary 2.31. If X is nonempty and compact and $f : X \rightarrow \mathbb{R}$ is continuous, then f attains its minimum and maximum.

Theorem 2.32. A subset K of a topological space is compact if and only if every collection of closed subsets of K having the finite intersection property has nonempty intersection.

Proof. Take complements in K . A family of closed sets has the finite intersection property exactly when the corresponding family of open complements has no finite subcover; its total intersection is empty exactly when those complements cover K . $\square$

Theorem 2.33. If $\mathcal{B}$ is a base for X , then $K \subseteq X$ is compact if and only if every cover of K by members of $\mathcal{B}$ has a finite subcover.

Proof. Only the converse needs proof. Refine each member of an arbitrary open cover by basic open subsets. A finite subcover from the resulting basic cover gives a finite subcover from the original cover. $\square$

Theorem 2.34 (Alexander's Subbase Theorem). If $\mathcal{S}$ is a subbase for the topology of X , then X is compact if and only if every cover of X by members of $\mathcal{S}$ has a finite subcover.

Proof. Only the reverse implication is nontrivial. Suppose every subbasic cover has a finite subcover, but X is not compact. By Zorn's lemma, there is a maximal open cover $\mathcal{C}$ of X having no finite subcover.

We claim that $\mathcal{C} \cap \mathcal{S}$ covers X . Otherwise choose $x \in X$ that belongs to no subbasic member of $\mathcal{C}$. Since $\mathcal{C}$ covers X , choose $U \in \mathcal{C}$ with $x \in U$. There are $S_1, \dots, S_n \in \mathcal{S}$ such that

$$x \in S_1 \cap \dots \cap S_n \subseteq U.$$

None of the S_i belongs to $\mathcal{C}$. By maximality, $\mathcal{C} \cup \{S_i\}$ has a finite subcover for each i ; hence $X = S_i \cup C_i$, where C_i is a finite union of members of $\mathcal{C}$. It follows that

$$X = \bigcap_{i=1}^n (S_i \cup C_i) \subseteq U \cup \bigcup_{i=1}^n C_i,$$

which is a finite subcover drawn from $\mathcal{C}$, a contradiction. Thus $\mathcal{C} \cap \mathcal{S}$ is a subbasic cover, so it has a finite subcover, again a contradiction. $\square$

Definition 2.35. A topological space is connected if its only clopen subsets are X and $\emptyset$. A subset is connected if it is connected with its subspace topology.

Theorem 2.36. Let X be a topological space.

- (i) The continuous image of a connected set is connected.
- (ii) If $\{E_i : i \in I\}$ is a family of connected subsets with $E_i \cap E_j \neq \emptyset$ for all i, j , then $\bigcup_i E_i$ is connected. If (E_n) is a sequence of connected sets with $E_n \cap E_{n+1} \neq \emptyset$, then $\bigcup_n E_n$ is connected.
- (iii) The union of two intersecting connected sets is connected.
- (iv) Every connected set is contained in a component; distinct components are disjoint; and the components cover X .
- (v) If C is connected and $C \subseteq Y \subseteq \text{cl } C$, then Y is connected.
- (vi) The closure of a connected set is connected, and every component is closed.

Proof. The proofs are the same as in metric spaces; they use only inverse images of open sets, relative topology, and the definition of closure. $\square$

Definition 2.37. A topological space X is locally connected if, for each $x \in X$ and each neighborhood G of x , there is a connected open neighborhood U of x with $U \subseteq G$.

Theorem 2.38. A topological space is locally connected if and only if it has a base consisting of connected open sets.

Proof. If X is locally connected, let $\mathcal{B}$ be the collection of all connected open subsets. Given $x \in G_1 \cap G_2$ with $G_1, G_2 \in \mathcal{B}$, local connectedness applied to the neighborhood $G_1 \cap G_2$ gives $B \in \mathcal{B}$ with $x \in B \subseteq G_1 \cap G_2$. Thus $\mathcal{B}$ is a base. The converse follows immediately from the definition of a base. $\square$

Theorem 2.39. If X is locally connected, then every component of X is both open and closed.

Proof. Components are closed in every topological space. If C is a component and $x \in C$, local connectedness gives a connected open neighborhood U of x . Since $U \cap C \neq \emptyset$, the set $U \cup C$ is connected, so maximality gives $U \subseteq C$. Therefore C is open. $\square$

Theorem 2.40. If X is locally connected and $f : X \rightarrow Z$ is continuous, open, and surjective, then Z is locally connected.

Proof. Let $z \in Z$ and let G be an open neighborhood of z . Choose $x \in X$ with $f(x) = z$. The set $f^{-1}(G)$ is an open neighborhood of x , so it contains a connected open neighborhood U of x . Then $f(U)$ is connected, open, contains z , and is contained in G . $\square$

Example 2.41. (a) A connected space need not be locally connected; the topologist's sine curve is a standard example.

(b) A locally connected space need not be connected; any discrete space with more than one point is an example.

Exercise 2.42. If X is compact, Z is Hausdorff, and $f : X \rightarrow Z$ is a continuous bijection, show that f is a homeomorphism.

Proof. If $F \subseteq X$ is closed, then F is compact, so $f(F)$ is compact and therefore closed in the Hausdorff space Z . Thus f is a closed bijection, and f^{-1} is continuous. $\square$

Exercise 2.43. Use Alexander's Subbase Theorem to prove that a finite product of compact spaces is compact.

Proof. Use the standard subbase

$$\{\pi_k^{-1}(U) : 1 \leq k \leq n, U \text{ open in } X_k\}.$$

Let $\mathcal{C}$ be a cover by such sets. For each k , let

$$\mathcal{C}_k = \{U \subseteq X_k : \pi_k^{-1}(U) \in \mathcal{C}\}.$$

At least one $\mathcal{C}_k$ covers X_k ; otherwise choose $x_k \notin \bigcup \mathcal{C}_k$ for every k , and the point $(x_1, \dots, x_n)$ would lie in no member of $\mathcal{C}$. Compactness of that factor supplies a finite subcover, whose inverse images form a finite subcover of the product. $\square$

Exercise 2.44. Give $\{0, 1\}$ the discrete topology. Show that X is connected if and only if every continuous function $X \rightarrow \{0, 1\}$ is constant.

Proof. For a continuous $f : X \rightarrow \{0, 1\}$, the sets $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen and partition X . Thus a nonconstant f gives a separation, and every separation gives such a nonconstant function. $\square$

Exercise 2.45. Assume that $X_1, \dots, X_n$ are nonempty. Show that the finite product $X_1 \times \dots \times X_n$ is locally connected if and only if every factor X_k is locally connected.

Proof. If the product is locally connected, each factor is locally connected because its projection is continuous, open, and surjective. Conversely, let $x = (x_1, \dots, x_n)$ belong to a basic neighborhood $G_1 \times \dots \times G_n$. Choose a connected open neighborhood U_k of x_k contained in G_k . The finite product $U_1 \times \dots \times U_n$ is connected and is an open neighborhood of x contained in the given basic neighborhood. $\square$

2.5 Pathwise connectedness

Definition 2.46. A path in X from p to q is a continuous function $f : [0, 1] \rightarrow X$ such that $f(0) = p$ and $f(1) = q$. A path with $p = q$ is a loop. A space is path connected if every pair of points can be joined by a path.

Theorem 2.47. If f is a path from a to b and g is a path from b to c , then

$$h(t) = \begin{cases} f(2t), & 0 \leq t \leq \frac{1}{2}, \\ g(2t - 1), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

is a path from a to c .

Proof. The two formulas agree at $t = 1/2$, so the Pasting Lemma gives continuity. $\square$

Theorem 2.48. Let X be a topological space.

- (a) Every path-connected space is connected.
- (b) A continuous image of a path-connected space is path connected; in particular, this holds for every continuous surjection.
- (c) If $\{E_i : i \in I\}$ is a collection of path-connected subsets with $E_i \cap E_j \neq \emptyset$ for all i, j , then $\bigcup_i E_i$ is path connected.
- (d) If (E_n) is a sequence of path-connected subsets with $E_n \cap E_{n+1} \neq \emptyset$, then $\bigcup_n E_n$ is path connected.

Proof. (a) If $X = A \cup B$ were a separation and a path joined a point of A to a point of B , its inverse images of A and B would separate $[0, 1]$.

(b) Choose preimages of the two target points and compose a path between them with the continuous map.

(c) Fix a common reference set E_{i_0} . Join each point to a point of E_{i_0} inside its own set, and concatenate with a path in E_{i_0} .

(d) Concatenate paths through the successive nonempty intersections; between any two fixed points only finitely many sets are involved. $\square$

Theorem 2.49. *Every path-connected subset is contained in a maximal path-connected subset, called a path component. Distinct path components are disjoint, and the path components cover the space.*

Proof. Use the union of all path-connected subsets containing a fixed point. All of these sets meet at that point, so their union is path connected and maximal. $\square$

Definition 2.50. A space X is locally path connected if, for each $x \in X$ and every neighborhood G of x , there is a path-connected open neighborhood U of x with $U \subseteq G$.

Equivalently, X has a base consisting of path-connected open sets.

Theorem 2.51. *If X is locally path connected, then every open connected subset of X is path connected.*

Proof. Let $G \subseteq X$ be open. Every path component of G is open in G : through each of its points there is a path-connected open neighborhood contained in G , and that neighborhood lies in the same path component. If G were connected and had more than one path component, one path component and the union of the others would form a separation. Hence G has one path component. $\square$

Corollary 2.52. *An open subset of $\mathbb{R}^q$ is connected if and only if it is path connected.*

Proof. Euclidean space is locally path connected because open balls are path connected. $\square$

Remark. Path connectedness does not imply local path connectedness, and local path connectedness does not imply path connectedness.

2.6 Infinite products

Definition 2.53. Let I be nonempty and let each X_i be nonempty. Their Cartesian product is

$$\prod_{i \in I} X_i = \left\{ x : I \rightarrow \bigcup_{i \in I} X_i : x(i) \in X_i \text{ for all } i \right\}.$$

We write x_i for $x(i)$. The coordinate projection is $\pi_i(x) = x_i$.

Theorem 2.54. *Let X be a set, let X_i be topological spaces, and let $f_i : X \rightarrow X_i$ be functions. The collection*

$$\mathcal{S} = \{f_i^{-1}(G) : i \in I, G \text{ open in } X_i\}$$

is a subbase for a topology $\mathcal{T}$ on X . This is the smallest topology making every f_i continuous. If the spaces X_i are Hausdorff and the functions separate points, then $\mathcal{T}$ is Hausdorff.

Proof. The subbase construction gives a topology. Every f_i is continuous by definition. Any topology making every f_i continuous must contain every subbasic set and hence must contain $\mathcal{T}$. If $x \neq y$ and $f_i(x) \neq f_i(y)$ for some i , disjoint neighborhoods of these two images have disjoint inverse images separating x and y . $\square$

Definition 2.55. The topology in the preceding theorem is the weak topology defined by the family $\{f_i : i \in I\}$.

Corollary 2.56. If Z is a topological space and $g : Z \rightarrow X$, where X has the weak topology defined by $\{f_i\}$, then g is continuous if and only if every $f_i \circ g$ is continuous.

Proof. One direction follows by composition. Conversely, the inverse image under g of a subbasic set $f_i^{-1}(G)$ is $(f_i \circ g)^{-1}(G)$, which is open. $\square$

Definition 2.57. For a family of topological spaces $\{X_i : i \in I\}$, the weak topology on $\prod_i X_i$ defined by the coordinate projections is the product topology. The sets $\pi_i^{-1}(G)$ form the standard subbase.

Theorem 2.58. Let $X = \prod_i X_i$ have the product topology.

- (a) Every projection $\pi_i : X \rightarrow X_i$ is open.
- (b) A map $g : Z \rightarrow X$ is continuous if and only if every coordinate map $\pi_i \circ g$ is continuous.

Proof. (a) The projection of a basic open set is open because all factors are nonempty. Projections preserve unions.

(b) This is the weak-topology criterion. $\square$

Theorem 2.59. Let (X_n, d_n) be metric spaces and put $X = \prod_{n=1}^{\infty} X_n$. Then the product topology is induced by the metric

$$D(x, y) = \sum_{n=1}^{\infty} 2^{-n} \frac{d_n(x_n, y_n)}{1 + d_n(x_n, y_n)}.$$

Proof. Each summand is a bounded metric after composing d_n with $t \mapsto t/(1+t)$, so the convergent series defines a metric.

Let G be a basic product neighborhood of x , restricting only the coordinates $n_1, \dots, n_k$, say $d_{n_j}(x_{n_j}, y_{n_j}) < r_j$. Choose $\varepsilon > 0$ so small that

$$2^{n_j} \varepsilon < \frac{r_j}{1 + r_j} \quad (1 \leq j \leq k).$$

Then $D(x, y) < \varepsilon$ implies all the required coordinate inequalities, so a D -ball lies in G .

Conversely, let $B_D(x; \varepsilon)$ be a metric ball. Choose N so that $\sum_{n > N} 2^{-n} < \varepsilon/2$. Choose $\delta > 0$ such that

$$\sum_{n=1}^N 2^{-n} \frac{\delta}{1 + \delta} < \frac{\varepsilon}{2}.$$

Then the basic product neighborhood

$$\bigcap_{n=1}^N \pi_n^{-1}(B_{d_n}(x_n; \delta))$$

is contained in $B_D(x; \varepsilon)$. Thus the two topologies agree. $\square$

Theorem 2.60 (Tychonoff's Theorem). *A product of nonempty topological spaces is compact if and only if every factor is compact.*

Proof. If the product is compact, every factor is compact as the continuous image under a surjective projection.

Conversely, suppose every X_i is compact. By Alexander's Subbase Theorem it is enough to consider a cover $\mathcal{C}$ by sets of the form $\pi_i^{-1}(G)$. For each i , let

$$\mathcal{C}_i = \{G \subseteq X_i : G \text{ is open and } \pi_i^{-1}(G) \in \mathcal{C}\}.$$

Some $\mathcal{C}_i$ must cover X_i . Otherwise choose $x_i \in X_i \setminus \bigcup \mathcal{C}_i$ for every i ; then $x = (x_i)$ lies in no member of $\mathcal{C}$. Compactness of that factor gives a finite subcover, whose inverse images form a finite subcover of the product. $\square$

Lemma 2.61. *Let $X = \prod_i X_i$, fix $a \in X$, and set*

$$D = \{x \in X : x_i = a_i \text{ for all but finitely many } i\}.$$

Then D is dense in X .

Proof. A nonempty basic open set restricts only finitely many coordinates. Choose a point satisfying those restrictions and set every unrestricted coordinate equal to the corresponding coordinate of a . The resulting point belongs to both the basic open set and D . $\square$

Theorem 2.62. *A product $\prod_i X_i$ is connected if and only if every factor X_i is connected.*

Proof. One direction follows from the projections. Conversely, fix $a \in X$. For each finite $J \subseteq I$, let

$$X_J = \{x \in X : x_i = a_i \text{ for } i \notin J\}.$$

The set X_J is homeomorphic to the finite product $\prod_{i \in J} X_i$, which is connected. All the sets X_J contain a , so their union D is connected. By the lemma, D is dense; therefore $X = \text{cl } D$ is connected. $\square$

Theorem 2.63. *A product $\prod_i X_i$ is path connected if and only if every factor is path connected.*

Proof. One direction follows from the projections. Conversely, for $x, y \in X$, choose a path $f_i : [0, 1] \rightarrow X_i$ from x_i to y_i for every i . Define $f(t) = (f_i(t))_{i \in I}$. Every coordinate $\pi_i \circ f = f_i$ is continuous, so f is continuous by the product criterion. $\square$

2.7 Nets

Definition 2.64. A directed set is a nonempty partially ordered set I such that, for every $i, j \in I$, there is a $k \in I$ with $i \leq k$ and $j \leq k$.

A net in a set X is a function $x : I \rightarrow X$ from a directed set; it is usually written $(x_i)_{i \in I}$.

A net in a topological space converges to x , written $x_i \rightarrow x$, if for every neighborhood G of x there is an i_0 such that $x_i \in G$ whenever $i \geq i_0$.

The net clusters at x , written $x_i \rightarrow_{\text{cl}} x$, if for every neighborhood G of x and every $j \in I$, there is an $i \geq j$ such that $x_i \in G$.

Theorem 2.65. (a) *Every convergent net clusters at its limit.*

(b) *In a Hausdorff space, a net has at most one limit.*

Proof. Part (a) is immediate. For (b), suppose $x_i \rightarrow x$ and $x_i \rightarrow y$ with $x \neq y$. Choose disjoint neighborhoods G of x and H of y . The net is eventually in each; choose an index larger than both corresponding indices to obtain a contradiction. $\square$

Theorem 2.66. *Let X and Z be topological spaces.*

(a) *A function $f : X \rightarrow Z$ is continuous at x if and only if $x_i \rightarrow x$ implies $f(x_i) \rightarrow f(x)$ for every net.*

(b) *If f is continuous at x and $x_i \rightarrow_{\text{cl}} x$, then $f(x_i) \rightarrow_{\text{cl}} f(x)$.*

(c) *A subset $F \subseteq X$ is closed if and only if every convergent net of points of F has its limit in F .*

Proof. The forward implications follow directly from the neighborhood definition. For the converse in (a), suppose f is not continuous at x . There is a neighborhood W of $f(x)$ such that no neighborhood of x is contained in $f^{-1}(W)$. Direct the neighborhoods of x by reverse inclusion, and for each neighborhood U choose $x_U \in U \setminus f^{-1}(W)$. Then $x_U \rightarrow x$, but $f(x_U)$ is never in W , a contradiction.

For (c), if F is not closed and $x \in \text{cl } F \setminus F$, direct the neighborhoods of x by reverse inclusion and choose $x_U \in U \cap F$. Then $x_U \rightarrow x$. $\square$

Theorem 2.67. *A topological space X is compact if and only if every net in X has a cluster point.*

Proof. Suppose first that every net has a cluster point. If an open cover $\mathcal{G}$ had no finite subcover, direct the finite subcollections $\mathcal{F} \subseteq \mathcal{G}$ by inclusion and choose

$$x_{\mathcal{F}} \in X \setminus \bigcup \mathcal{F}.$$

Let x be a cluster point and choose $G \in \mathcal{G}$ with $x \in G$. Taking $\mathcal{F}_0 = \{G\}$, the clustering property gives $\mathcal{F} \supseteq \mathcal{F}_0$ with $x_{\mathcal{F}} \in G$, contradicting the choice of $x_{\mathcal{F}}$.

Conversely, let (x_i) be a net in compact X , and put

$$F_j = \text{cl}\{x_i : i \geq j\}.$$

The family (F_j) has the finite intersection property, so choose $x \in \bigcap_j F_j$. Every neighborhood of x meets every tail of the net; hence x is a cluster point. $\square$

Theorem 2.68. *If X is compact and a net (x_i) has a unique cluster point x , then $x_i \rightarrow x$.*

Proof. If the net did not converge to x , there would be a neighborhood U of x such that every tail contains a point in the closed set $X \setminus U$. The closed sets

$$(X \setminus U) \cap \text{cl}\{x_j : j \geq i\}$$

have the finite intersection property. Compactness would give a point $y \in X \setminus U$ belonging to all of them, and y would be another cluster point. $\square$

Exercise 2.69. Let $\mathcal{S}$ be a subbase for X .

(a) Show that $x_i \rightarrow x$ if and only if, for every $S \in \mathcal{S}$ with $x \in S$, the net is eventually in S .

(b) Find a subbase, a net, and a point x such that the net is frequently in every subbasic neighborhood of x but does not cluster at x .

Proof. (a) The forward implication is immediate. Conversely, a basic neighborhood of x is a finite intersection $S_1 \cap \cdots \cap S_n$ of subbasic neighborhoods. Choose a common upper bound for the finitely many indices after which the net lies in each S_k .

(b) Let $X = \{1, 2, 3\}$ with the discrete topology and

$$\mathcal{S} = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$

This is a subbase. Let $x = 1$ and let $x_n = 2$ for even n and $x_n = 3$ for odd n . The net is frequently in each subbasic neighborhood of 1, but it never enters the neighborhood $\{1\}$, so it does not cluster at 1. $\square$

Exercise 2.70. Let $X = \prod_{\alpha \in A} X^\alpha$, let (x_i) be a net in X , and write $x_i = (x_i^\alpha)$ and $x = (x^\alpha)$.

(a) Show that $x_i \rightarrow x$ if and only if $x_i^\alpha \rightarrow x^\alpha$ for every α .

(b) Give a sequence in $\mathbb{R}^2$ whose two coordinate sequences cluster at x^1 and x^2 , respectively, but which does not cluster at (x^1, x^2) .

Proof. (a) This is the product continuity criterion applied to the coordinate projections, or directly the subbase criterion for nets.

(b) Define

$$x_n = \begin{cases} (1/n, 0), & n \text{ even,} \\ (1, 1 - 1/n), & n \text{ odd.} \end{cases}$$

The first coordinate clusters at 0, the second at 1, but the sequence in $\mathbb{R}^2$ does not cluster at $(0, 1)$. $\square$

2.8 Quotient spaces

Definition 2.71. An equivalence relation $\sim$ on a set X is reflexive, symmetric, and transitive.

Theorem 2.72. *The equivalence classes of an equivalence relation form a partition of X . Conversely, every partition of X defines an equivalence relation by declaring two points equivalent when they lie in the same member of the partition.*

Definition 2.73. The set of equivalence classes is denoted $X/\sim$. The natural map $q : X \rightarrow X/\sim$ sends each point to its equivalence class and is called the quotient map.

Theorem 2.74. *Let X be a topological space with an equivalence relation $\sim$. The collection*

$$\mathcal{U} = \{U \subseteq X/\sim : q^{-1}(U) \text{ is open in } X\}$$

is a topology on $X/\sim$, called the quotient topology. The map q is continuous and surjective. The quotient need not be Hausdorff.

Proof. Inverse images preserve the empty set, the whole set, arbitrary unions, and finite intersections, so $\mathcal{U}$ is a topology. Continuity of q is built into its definition. $\square$

Theorem 2.75. *Let $q : X \rightarrow X/\sim$ be a quotient map and let $f : X/\sim \rightarrow Z$. Then f is continuous if and only if $f \circ q : X \rightarrow Z$ is continuous.*

Proof. One direction follows by composition. Conversely, if $U \subseteq Z$ is open, then

$$q^{-1}(f^{-1}(U)) = (f \circ q)^{-1}(U)$$

is open in X . By the definition of the quotient topology, $f^{-1}(U)$ is open. $\square$

Exercise 2.76. Let $q : X \rightarrow X/\sim$ be an open quotient map. If X is locally connected, show that $X/\sim$ is locally connected.

Proof. The map q is continuous, open, and surjective, so the result follows from the theorem on open continuous images of locally connected spaces. $\square$

Exercise 2.77. If X is path connected and $\sim$ is any equivalence relation, show that $X/\sim$ is path connected.

Proof. The quotient map is continuous and surjective, and continuous images of path-connected spaces are path connected. $\square$

Exercise 2.78. A semimetric (or pseudometric) on X is a function $\rho : X \times X \rightarrow [0, \infty)$ satisfying symmetry, the triangle inequality, and $\rho(x, x) = 0$, but distinct points may have distance zero. Give X the topology generated by the semimetric balls. Define $x \sim y$ when $\rho(x, y) = 0$.

- (a) Show that $\sim$ is an equivalence relation.
- (b) Show that

$$d(q(x), q(y)) = \rho(x, y)$$

is a well-defined metric on $X/\sim$.

- (c) Show that this metric induces the quotient topology.

Proof. (a) Reflexivity and symmetry are immediate. If $\rho(x, y) = 0$ and $\rho(y, z) = 0$, then the triangle inequality gives $\rho(x, z) = 0$.

- (b) If $x \sim x'$ and $y \sim y'$, then

$$\rho(x, y) \leq \rho(x, x') + \rho(x', y') + \rho(y', y) = \rho(x', y'),$$

and the reverse inequality follows symmetrically. Thus d is well defined. The metric axioms follow from those of ρ , and positive definiteness holds because distinct equivalence classes have positive semidistance.

- (c) For every $x \in X$ and $r > 0$,

$$q^{-1}(B_d(q(x); r)) = B_\rho(x; r).$$

Thus metric balls in the quotient are quotient-open. Conversely, if $U \subseteq X/\sim$ is quotient-open and $q(x) \in U$, then $q^{-1}(U)$ is an open neighborhood of x , so some $B_\rho(x; r)$ lies in $q^{-1}(U)$. Hence $B_d(q(x); r) \subseteq U$, proving that U is metric-open. $\square$

3 Continuous real-valued functions

3.1 Convergence of functions

Definition 3.1. For a topological space X , let $C(X)$ denote the vector space of all continuous functions from X to $\mathbb{R}$.

A function $f : X \rightarrow \mathbb{R}$ is bounded if there is a constant M such that $|f(x)| \leq M$ for every $x \in X$. Let $C_b(X)$ denote the space of all bounded continuous functions from X to $\mathbb{R}$.

For $f \in C_b(X)$, write

$$\|f\|_\infty = \sup_{x \in X} |f(x)|,$$

with the convention that the supremum is 0 when $X = \emptyset$. For $f, g \in C_b(X)$, define

$$\rho(f, g) = \|f - g\|_\infty.$$

Theorem 3.2. *The function ρ is a metric on $C_b(X)$.*

Proof. Nonnegativity and symmetry are immediate. If $\rho(f, g) = 0$, then $|f(x) - g(x)| = 0$ for every x , so $f = g$. Finally, for every $x \in X$,

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|.$$

Taking suprema gives

$$\rho(f, h) \leq \rho(f, g) + \rho(g, h).$$

□

Definition 3.3. Let $f_n, f : X \rightarrow \mathbb{R}$ be bounded functions. The sequence (f_n) converges uniformly to f if, for every $\varepsilon > 0$, there is an N such that

$$|f_n(x) - f(x)| < \varepsilon$$

for every $x \in X$ and every $n \geq N$.

The sequence (f_n) is uniformly Cauchy if, for every $\varepsilon > 0$, there is an N such that

$$|f_n(x) - f_m(x)| < \varepsilon$$

for every $x \in X$ and all $m, n \geq N$.

Theorem 3.4. *Let X be a topological space and let (f_n) be a sequence in $C_b(X)$. Then $f_n \rightarrow f$ in the metric ρ if and only if $f_n \rightarrow f$ uniformly. Moreover, (f_n) is Cauchy in $(C_b(X), \rho)$ if and only if it is uniformly Cauchy.*

Proof. Both assertions follow immediately from the definition of the supremum metric. □

Theorem 3.5. *For every topological space X , the metric space $(C_b(X), \rho)$ is complete.*

Proof. Let (f_n) be Cauchy in $C_b(X)$. For each $x \in X$, the real sequence $(f_n(x))$ is Cauchy, so define

$$f(x) = \lim_{n \rightarrow \infty} f_n(x).$$

Given $\varepsilon > 0$, choose N such that $\rho(f_n, f_m) < \varepsilon/2$ whenever $m, n \geq N$. Letting $m \rightarrow \infty$ gives

$$|f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon$$

for every $x \in X$ and $n \geq N$. Hence $f_n \rightarrow f$ uniformly.

Taking $\varepsilon = 1$ and fixing $n \geq N$ shows

$$|f(x)| \leq |f(x) - f_n(x)| + |f_n(x)| \leq 1 + \|f_n\|_\infty,$$

so f is bounded. A uniform limit of continuous functions is continuous: if $x \in X$ and $\varepsilon > 0$, choose n with $\rho(f_n, f) < \varepsilon/3$ and then a neighborhood U of x such that $|f_n(y) - f_n(x)| < \varepsilon/3$ for $y \in U$. For $y \in U$,

$$|f(y) - f(x)| < \varepsilon.$$

Thus $f \in C_b(X)$ and $f_n \rightarrow f$ in ρ . □

Theorem 3.6 (Dini's theorem). *Let X be compact. Suppose (f_n) is an increasing sequence in $C(X)$ and $f \in C(X)$ satisfies $f_n(x) \rightarrow f(x)$ for every $x \in X$. Then $f_n \rightarrow f$ uniformly on X .*

Proof. Fix $\varepsilon > 0$ and put

$$U_n = \{x \in X : f(x) - f_n(x) < \varepsilon\}.$$

Each U_n is open, $U_n \subseteq U_{n+1}$, and pointwise convergence implies that $\bigcup_{n=1}^{\infty} U_n = X$. Compactness gives $X = U_{n_1} \cup \dots \cup U_{n_k}$. If $N = \max(n_1, \dots, n_k)$, then $U_N = X$. Since $f_n \leq f$ for every n , we have

$$0 \leq f(x) - f_N(x) < \varepsilon$$

for all $x \in X$. The same estimate holds for every $n \geq N$. $\square$

3.2 Separation properties

Definition 3.7. A topological space X is regular if, whenever $F \subseteq X$ is closed and $x \notin F$, there are disjoint open sets U, V such that $x \in U$ and $F \subseteq V$.

Theorem 3.8. For a topological space X , the following are equivalent.

- (a) X is regular.
- (b) If G is open and $x \in G$, then there is an open set U such that

$$x \in U \subseteq \text{cl } U \subseteq G.$$

- (c) If F is closed and $x \notin F$, then there is an open set V such that

$$F \subseteq V \quad \text{and} \quad x \notin \text{cl } V.$$

Proof. Assume (a). Apply regularity to x and $X \setminus G$. If U is the open neighborhood of x and V is the open set containing $X \setminus G$, then

$$\text{cl } U \subseteq X \setminus V \subseteq G,$$

which proves (b).

Assume (b), put $G = X \setminus F$, and choose U as in (b). Then $V = X \setminus \text{cl } U$ is open, contains F , and $x \notin \text{cl } V$ because $U \cap V = \emptyset$.

Finally, assume (c). Given F and $x \notin F$, choose V as in (c) and put $U = X \setminus \text{cl } V$. Then U and V are disjoint open sets containing x and F , respectively. $\square$

Theorem 3.9. (a) Every subspace of a regular space is regular.

(b) Let $X = \prod_{i \in I} X_i$, where every factor is nonempty. Then X is regular if and only if every X_i is regular.

Proof. (a) Let $Y \subseteq X$, let $y \in Y$, and let G be a neighborhood of y in Y . Write $G = Y \cap H$ with H open in X . Choose an open $U \subseteq X$ such that

$$y \in U \subseteq \text{cl}_X U \subseteq H.$$

Then

$$y \in Y \cap U \subseteq \text{cl}_Y(Y \cap U) \subseteq Y \cap \text{cl}_X U \subseteq G.$$

Thus Y is regular.

(b) If the product is regular, each factor is homeomorphic to a subspace obtained by fixing all the other coordinates, so (a) applies.

Conversely, suppose every factor is regular. Let $x \in X$ and let G be an open neighborhood of x . Choose a basic neighborhood

$$B = \bigcap_{k=1}^n \pi_{i_k}^{-1}(G_{i_k})$$

with $x \in B \subseteq G$. For each k , choose an open set U_{i_k} such that

$$x_{i_k} \in U_{i_k} \subseteq \text{cl } U_{i_k} \subseteq G_{i_k}.$$

Then

$$U = \bigcap_{k=1}^n \pi_{i_k}^{-1}(U_{i_k})$$

is open, contains x , and satisfies

$$\text{cl } U \subseteq \bigcap_{k=1}^n \pi_{i_k}^{-1}(\text{cl } U_{i_k}) \subseteq B \subseteq G.$$

□

Theorem 3.10. *Every compact Hausdorff space and every metric space is regular.*

Proof. Let X be compact Hausdorff, let F be closed, and let $x \notin F$. For each $y \in F$, choose disjoint open sets U_y, V_y with $x \in U_y$ and $y \in V_y$. A finite number of the sets V_y cover the compact set F . The intersection of the corresponding U_y is then an open neighborhood of x disjoint from the union of those V_y .

Now let (X, d) be metric. The case $F = \emptyset$ is immediate. Otherwise, for closed F and $x \notin F$, the function

$$h(y) = \frac{d(y, F)}{d(y, F) + d(y, x)}$$

is continuous, equals 0 on F , and equals 1 at x . The inverse images of $(-1/3, 1/3)$ and $(2/3, 4/3)$ give disjoint open neighborhoods of F and x . □

Definition 3.11. A topological space X is completely regular if, whenever $F \subseteq X$ is closed and $x \notin F$, there is a continuous function $f : X \rightarrow \mathbb{R}$ such that $f(x) = 1$ and $f|_F = 0$.

Theorem 3.12. *A space X is completely regular if and only if the separating function in the definition can always be chosen with values in $[0, 1]$.*

Proof. Only one direction requires proof. Given a real-valued separating function f , the function

$$h(x) = \min\{1, \max\{0, f(x)\}\}$$

is continuous, takes values in $[0, 1]$, equals 1 at the chosen point, and vanishes on the closed set. □

Theorem 3.13. *Every subspace of a completely regular space is completely regular.*

Proof. Let $Y \subseteq X$, let F be closed in Y , and let $y \in Y \setminus F$. There is a closed set $C \subseteq X$ such that $F = Y \cap C$. Since $y \notin C$, choose $f : X \rightarrow [0, 1]$ with $f(y) = 1$ and $f|_C = 0$. Then $f|_Y$ separates y from F . □

Lemma 3.14. *Let $\mathcal{T}$ be the weak topology on X generated by maps $f_i : X \rightarrow X_i$, $i \in I$. A net (x_α) converges to x in $(X, \mathcal{T})$ if and only if*

$$f_i(x_\alpha) \longrightarrow f_i(x)$$

for every $i \in I$.

Proof. The forward implication follows from continuity of the maps f_i . Conversely, every basic neighborhood of x has the form

$$\bigcap_{k=1}^n f_{i_k}^{-1}(G_k),$$

where G_k is a neighborhood of $f_{i_k}(x)$. Coordinate convergence makes the net eventually belong to each of the finitely many inverse images, and therefore to their intersection. $\square$

Theorem 3.15. *For a topological space X , the following are equivalent.*

- (a) *X is completely regular.*
- (b) *The topology of X is the weak topology generated by $C(X)$.*
- (c) *The topology of X is the weak topology generated by $C_b(X)$.*
- (d) *A net (x_α) converges to x in X if and only if*

$$f(x_\alpha) \longrightarrow f(x)$$

for every continuous $f : X \rightarrow [0, 1]$.

Proof. Assume (a). The weak topology generated by $C(X)$ is no finer than the given topology. To prove the reverse inclusion, let G be open and let $x \in G$. Complete regularity gives $f : X \rightarrow [0, 1]$ such that $f(x) = 1$ and $f = 0$ on $X \setminus G$. Then

$$x \in f^{-1}((1/2, \infty)) \subseteq G,$$

so G is weakly open. This proves (b).

The weak topologies generated by $C(X)$ and $C_b(X)$ coincide. Indeed, $C_b(X) \subseteq C(X)$, while for every $f \in C(X)$ the bounded function

$$h = \frac{f}{1 + |f|}$$

belongs to $C_b(X)$ and

$$f = \frac{h}{1 - |h|}.$$

Thus every member of $C(X)$ is continuous for the weak topology generated by $C_b(X)$. Hence (b) and (c) are equivalent.

If (c) holds, the weak-topology lemma gives (d), since bounded real-valued functions may be continuously rescaled into $(-1, 1)$ and $[0, 1]$ -valued functions already generate the same topology.

Finally, assume (d). Let $\mathcal{T}_0$ be the weak topology generated by all continuous maps from X to $[0, 1]$. By the lemma, convergence in $\mathcal{T}_0$ is exactly coordinatewise convergence under these maps. Condition (d) says that $\mathcal{T}_0$ and the original topology have the same convergent nets, and therefore the same open sets.

Let F be closed and $x \notin F$. Since $X \setminus F$ is $\mathcal{T}_0$ -open, there are continuous functions $f_1, \dots, f_n : X \rightarrow [0, 1]$ and open sets $O_k \subseteq [0, 1]$ such that

$$x \in \bigcap_{k=1}^n f_k^{-1}(O_k) \subseteq X \setminus F.$$

For each k , choose a continuous $u_k : [0, 1] \rightarrow [0, 1]$ that equals 1 at $f_k(x)$ and vanishes outside O_k . Then

$$h = \prod_{k=1}^n (u_k \circ f_k)$$

is continuous, satisfies $h(x) = 1$, and vanishes on F . Thus X is completely regular. $\square$

Theorem 3.16. *Let $\{X_i : i \in I\}$ be a collection of nonempty topological spaces. Then $\prod_{i \in I} X_i$ is completely regular if and only if every X_i is completely regular.*

Proof. If the product is completely regular, each factor is homeomorphic to a subspace obtained by fixing all other coordinates, so the hereditary theorem applies.

Conversely, let $X = \prod_i X_i$, let $F \subseteq X$ be closed, and let $x \notin F$. Choose a basic neighborhood

$$N = \bigcap_{k=1}^n \pi_{i_k}^{-1}(G_{i_k})$$

with $x \in N \subseteq X \setminus F$. For each k , complete regularity gives a continuous $u_k : X_{i_k} \rightarrow [0, 1]$ such that

$$u_k(x_{i_k}) = 1 \quad \text{and} \quad u_k = 0 \text{ on } X_{i_k} \setminus G_{i_k}.$$

The function

$$u(y) = \prod_{k=1}^n u_k(y_{i_k})$$

is continuous, equals 1 at x , and vanishes on F because every point of F lies outside N . $\square$

3.3 Normal spaces

Definition 3.17. A topological space X is normal if, whenever A and B are disjoint closed subsets of X , there are disjoint open sets U, V such that $A \subseteq U$ and $B \subseteq V$.

Theorem 3.18. *For a topological space X , the following are equivalent.*

- (a) X is normal.
- (b) If A is closed, G is open, and $A \subseteq G$, then there is an open set U such that

$$A \subseteq U \subseteq \text{cl } U \subseteq G.$$

- (c) If A and B are disjoint closed sets, then there is an open set V such that

$$B \subseteq V \quad \text{and} \quad A \cap \text{cl } V = \emptyset.$$

Proof. Assume (a). Separate A and $X \setminus G$ by disjoint open sets U and V . Then

$$A \subseteq U \subseteq \text{cl } U \subseteq X \setminus V \subseteq G,$$

which proves (b).

Assume (b). Apply it to $B \subseteq X \setminus A$ and choose V with

$$B \subseteq V \subseteq \text{cl } V \subseteq X \setminus A.$$

This proves (c).

Assume (c). Choose V as stated and put $U = X \setminus \text{cl } V$. Then U and V are disjoint open sets containing A and B , respectively. $\square$

Theorem 3.19. (a) *Every metric space is normal.*

(b) *Every compact Hausdorff space is normal.*

(c) *Every closed subspace of a normal space is normal.*

Proof. (a) If one of A and B is empty, the conclusion is immediate. Otherwise, for disjoint closed sets A and B in a metric space, the function

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}$$

is continuous, equals 0 on A , and equals 1 on B . Suitable inverse images of disjoint intervals separate A and B .

(b) Let A and B be disjoint closed subsets of a compact Hausdorff space. For each $a \in A$, regularity gives an open neighborhood U_a of a whose closure is disjoint from B . A finite number of these sets cover A . Their union U satisfies $A \subseteq U$ and $B \cap \text{cl } U = \emptyset$, so the preceding characterization proves normality.

(c) Let Y be closed in X . If A and B are closed in Y , then they are closed in X . Disjoint open neighborhoods of A and B in X restrict to disjoint open neighborhoods in Y . $\square$

The following proof uses the dyadic rationals

$$D = \left\{ \frac{k}{2^n} : n \geq 0, 0 \leq k \leq 2^n \right\} \subseteq [0, 1].$$

Theorem 3.20 (Urysohn's lemma). *If X is normal and $A, B \subseteq X$ are disjoint closed sets, then there is a continuous function $f : X \rightarrow [0, 1]$ such that*

$$f|_A = 0 \quad \text{and} \quad f|_B = 1.$$

Proof. Set $U_1 = X \setminus B$. By normality, choose an open set U_0 such that

$$A \subseteq U_0 \subseteq \text{cl } U_0 \subseteq U_1.$$

Inductively over the dyadic levels, choose the remaining open sets U_r , $r \in D$, so that

$$\text{cl } U_r \subseteq U_s \quad (r < s).$$

For the induction step, insert a new dyadic number r between its two adjacent previously constructed dyadics $p < r < q$ and choose

$$\text{cl } U_p \subseteq U_r \subseteq \text{cl } U_r \subseteq U_q.$$

Define

$$f(x) = \inf(\{r \in D : x \in U_r\} \cup \{1\}).$$

Then $f = 0$ on A . If $x \in B$, then $x \notin U_r$ for every $r \in D$, so $f(x) = 1$.

For $0 < a \leq 1$,

$$\{x : f(x) < a\} = \bigcup_{\substack{r \in D \\ r < a}} U_r,$$

and for $0 \leq a < 1$,

$$\{x : f(x) > a\} = \bigcup_{\substack{r \in D \\ r > a}} (X \setminus \text{cl } U_r).$$

These sets are open. For values of a outside the indicated ranges, the corresponding inverse image is either X or $\emptyset$. Hence f is continuous. $\square$

Corollary 3.21. *If X is normal, A and B are disjoint closed subsets, and $\alpha < \beta$, then there is a continuous function $f : X \rightarrow [\alpha, \beta]$ such that $f|_A = \alpha$ and $f|_B = \beta$.*

Proof. If u is supplied by Urysohn's lemma, take

$$f = \alpha + (\beta - \alpha)u.$$

$\square$

Theorem 3.22. *Every normal Hausdorff space is completely regular.*

Proof. Given a closed set F and $x \notin F$, apply Urysohn's lemma to the disjoint closed sets F and $\{x\}$. $\square$

Lemma 3.23 (Tietze approximation lemma). *Let X be normal, let $C \subseteq X$ be closed, and let $\gamma \geq 0$. Suppose $f : C \rightarrow \mathbb{R}$ is continuous and $|f(c)| \leq \gamma$ for all $c \in C$. Then there is a continuous function $g : X \rightarrow \mathbb{R}$ such that*

$$|g(x)| \leq \frac{\gamma}{3} \quad (x \in X)$$

and

$$|f(c) - g(c)| \leq \frac{2\gamma}{3} \quad (c \in C).$$

Proof. If $\gamma = 0$, take $g \equiv 0$. Assume henceforth that $\gamma > 0$. The sets

$$A = \left\{c \in C : f(c) \geq \frac{\gamma}{3}\right\}, \quad B = \left\{c \in C : f(c) \leq -\frac{\gamma}{3}\right\}$$

are disjoint and closed in X . By the preceding corollary, choose $g : X \rightarrow [-\gamma/3, \gamma/3]$ such that $g = \gamma/3$ on A and $g = -\gamma/3$ on B . On $A \cup B$ the error is at most $2\gamma/3$; outside $A \cup B$, both $|f|$ and $|g|$ are at most $\gamma/3$, so the same estimate holds. $\square$

Theorem 3.24 (Tietze extension theorem). *Let X be normal, let $C \subseteq X$ be closed, and let $\alpha \leq \beta$. If $f : C \rightarrow [\alpha, \beta]$ is continuous, then there is a continuous function $F : X \rightarrow [\alpha, \beta]$ such that $F|_C = f$.*

Proof. If $\alpha = \beta$, the constant function $F \equiv \alpha$ works. Otherwise, affine rescaling reduces the problem to a function $f : C \rightarrow [0, 1]$. Put $r_0 = f$. Apply the approximation lemma successively to obtain continuous functions $g_n : X \rightarrow \mathbb{R}$ such that, on C ,

$$r_n = f - \sum_{k=1}^n g_k, \quad |r_n| \leq \left(\frac{2}{3}\right)^n,$$

and, on X ,

$$|g_n| \leq \frac{1}{3} \left(\frac{2}{3} \right)^{n-1}.$$

The Weierstrass M -test shows that

$$g = \sum_{n=1}^{\infty} g_n$$

converges uniformly and is continuous. Since $r_n \rightarrow 0$ on C , we have $g|_C = f$. Moreover, $|g| \leq 1$ on X . Therefore

$$F(x) = \min\{1, \max\{0, g(x)\}\}$$

is a continuous $[0, 1]$ -valued extension of the rescaled function. Rescaling back gives the result for $[\alpha, \beta]$. $\square$

Theorem 3.25 (Finite partition of unity). *Let X be normal and let $G_1, \dots, G_n$ be a finite open cover of X . Then there are continuous functions $\phi_1, \dots, \phi_n : X \rightarrow [0, 1]$ such that*

$$\phi_k = 0 \text{ on } X \setminus G_k$$

and

$$\sum_{k=1}^n \phi_k = 1.$$

Proof. First shrink the cover. Construct open sets $V_1, \dots, V_n$ inductively so that

$$\text{cl } V_k \subseteq G_k$$

and $V_1, \dots, V_k, G_{k+1}, \dots, G_n$ still cover X . At the k th step, the closed set

$$F_k = X \setminus (V_1 \cup \dots \cup V_{k-1} \cup G_{k+1} \cup \dots \cup G_n)$$

is contained in G_k . Normality gives an open V_k with

$$F_k \subseteq V_k \subseteq \text{cl } V_k \subseteq G_k.$$

Thus $V_1, \dots, V_n$ cover X .

By Urysohn's lemma, choose $\psi_k : X \rightarrow [0, 1]$ such that $\psi_k = 1$ on $\text{cl } V_k$ and $\psi_k = 0$ on $X \setminus G_k$. The function $s = \sum_{k=1}^n \psi_k$ is continuous and satisfies $s \geq 1$. Set

$$\phi_k = \frac{\psi_k}{s}.$$

These functions have all the required properties. $\square$

The functions in the preceding theorem are called a partition of unity subordinate to the finite open cover $\{G_1, \dots, G_n\}$.

Corollary 3.26. *Let K be a closed subset of a normal space X , and let $G_1, \dots, G_n$ be open sets covering K . Then there are continuous functions $\phi_1, \dots, \phi_n : X \rightarrow [0, 1]$ such that*

- (a) $\phi_k = 0$ on $X \setminus G_k$;
- (b) $\sum_{k=1}^n \phi_k = 1$ on K ;
- (c) $\sum_{k=1}^n \phi_k \leq 1$ on X .

Proof. Apply the finite partition theorem to the open cover

$$G_1, \dots, G_n, X \setminus K$$

and discard the last function. $\square$

3.4 The Stone–Čech compactification

Theorem 3.27 (Stone–Čech compactification). *Let X be completely regular. There are a compact Hausdorff space βX and a homeomorphism τ from X onto a dense subspace of βX such that every bounded continuous function $f : X \rightarrow \mathbb{R}$ has a continuous extension $f^\beta : \beta X \rightarrow \mathbb{R}$ satisfying*

$$f^\beta \circ \tau = f.$$

This compactification is unique up to a homeomorphism that fixes the embedded copy of X .

Proof. Let

$$\mathcal{F} = C(X, [0, 1]) \quad \text{and} \quad \Omega = [0, 1]^\mathcal{F}.$$

The space Ω is compact by Tychonoff's theorem. Define the evaluation map

$$\tau : X \longrightarrow \Omega, \quad \tau(x)_f = f(x).$$

Continuous $[0, 1]$ -valued functions separate distinct points of a completely regular Hausdorff space, so τ is injective. The topology of X is the weak topology generated by these functions, so τ is a homeomorphism onto its image. Put

$$\beta X = \text{cl}_\Omega \tau(X).$$

Then βX is compact Hausdorff and $\tau(X)$ is dense in it.

For $f \in \mathcal{F}$, the coordinate projection π_f restricted to βX extends f . Every bounded real-valued continuous function can be rescaled into $[0, 1]$, extended by a coordinate projection, and then rescaled back.

For uniqueness, suppose Z is another compact Hausdorff space with an embedding $\sigma : X \rightarrow Z$ having dense image and the same extension property. For each $f \in \mathcal{F}$, let f^Z be its extension to Z , and define

$$w : Z \longrightarrow \Omega, \quad w(z)_f = f^Z(z).$$

The coordinate maps of w are continuous, so w is continuous, and

$$w \circ \sigma = \tau.$$

The map w is injective. Indeed, if $z \neq z'$, normality of the compact Hausdorff space Z gives $h \in C(Z, [0, 1])$ with $h(z) \neq h(z')$. If $f = h \circ \sigma$, then h and f^Z agree on the dense set $\sigma(X)$, so they agree on all of Z ; hence the f -coordinate separates z and z' .

Since $w(Z)$ is compact, it is closed in Ω and contains $\tau(X)$, so $\beta X \subseteq w(Z)$. Conversely, density of $\sigma(X)$ and continuity of w give

$$w(Z) \subseteq \text{cl } w(\sigma(X)) = \text{cl } \tau(X) = \beta X.$$

Thus w is a continuous bijection from compact Z onto Hausdorff βX , and therefore a homeomorphism. $\square$

Remark. The Stone–Čech compactification can be much larger than a familiar compactification. For example, $(0, 1]$ is dense in $[0, 1]$, but the bounded continuous function $t \mapsto \sin(1/t)$ on $(0, 1]$ has no continuous extension to 0. Consequently, $[0, 1]$ is not the Stone–Čech compactification of $(0, 1]$.

Theorem 3.28 (Universal property). *Let X be completely regular and let Z be compact Hausdorff. Every continuous map $\eta : X \rightarrow Z$ has a unique continuous extension*

$$\eta^\beta : \beta X \rightarrow Z$$

satisfying $\eta^\beta \circ \tau = \eta$.

Proof. Let $\mathcal{H} = C(Z, [0, 1])$ and use the evaluation embedding

$$j : Z \longrightarrow [0, 1]^\mathcal{H}, \quad j(z)_h = h(z).$$

For every $h \in \mathcal{H}$, the function $h \circ \eta : X \rightarrow [0, 1]$ has a Stone–Čech extension $(h \circ \eta)^\beta$. Define

$$H : \beta X \longrightarrow [0, 1]^\mathcal{H}, \quad H(\xi)_h = (h \circ \eta)^\beta(\xi).$$

The map H is continuous and

$$H \circ \tau = j \circ \eta.$$

Since $j(Z)$ is compact and therefore closed in the product,

$$H(\beta X) \subseteq \text{cl } H(\tau(X)) \subseteq j(Z).$$

Hence

$$\eta^\beta = j^{-1} \circ H$$

is well defined and continuous. It extends η . Uniqueness follows because two continuous maps into the Hausdorff space Z that agree on the dense set $\tau(X)$ agree everywhere. $\square$

3.5 Locally compact spaces

Definition 3.29. A Hausdorff space X is locally compact if, for every $x \in X$ and every neighborhood G of x , there is a neighborhood U of x such that $\text{cl } U$ is compact and

$$\text{cl } U \subseteq G.$$

Theorem 3.30. *Let all spaces below be Hausdorff.*

(a) *A space X is locally compact if and only if every point of X has at least one neighborhood with compact closure.*

(b) *Every open or closed subspace of a locally compact space is locally compact.*

(c) *Let $X = \prod_{i \in I} X_i$, where every factor is nonempty. Then X is locally compact if and only if every X_i is locally compact and all but finitely many X_i are compact.*

(d) *A metric space (X, d) is locally compact if and only if, for every $x \in X$, there is an $r > 0$ such that $\text{cl } B(x; r)$ is compact.*

Proof. (a) One implication is immediate. Conversely, suppose x has a neighborhood G whose closure K is compact, and let O be any neighborhood of x . The compact Hausdorff space K is regular. In the relative open set $K \cap G \cap O$, choose a relative open set V such that

$$x \in V \subseteq \text{cl}_K V \subseteq K \cap G \cap O.$$

Write $V = K \cap W$ with W open in X , and put $U = G \cap W$. Then U is an open neighborhood of x and

$$\text{cl}_X U \subseteq \text{cl}_K V \subseteq O.$$

Its closure is a closed subset of the compact space K .

(b) Let E be closed and $x \in E$. A neighborhood G of x in X with compact closure gives the relative neighborhood $E \cap G$, whose closure in E is compact. Part (a) applies. If E is open, then every open neighborhood O of x in E is also open in X . Local compactness in X supplies a smaller neighborhood whose compact closure is contained in O , and hence in E .

(c) Suppose the product is locally compact. Each factor is homeomorphic to a closed slice of the product, so each factor is locally compact by (b). Choose $x \in X$ and a neighborhood U of x with compact closure. There is a basic neighborhood

$$B = \prod_{i \in I} B_i$$

with $x \in B \subseteq U$ and $B_i = X_i$ except for finitely many i . If $B_i = X_i$, then

$$X_i = \pi_i(B) \subseteq \pi_i(\text{cl } U) \subseteq X_i,$$

so $X_i = \pi_i(\text{cl } U)$ is compact. Hence only finitely many factors can be noncompact.

Conversely, let F be the finite set of noncompact factors. For each $i \in F$, choose a neighborhood U_i of x_i with compact closure. Then

$$U = \prod_{i \in F} U_i \times \prod_{i \notin F} X_i$$

is a neighborhood of x , and its closure is a product of compact spaces, hence compact by Tychonoff's theorem.

(d) If X is locally compact, choose a neighborhood G of x with compact closure and then an $r > 0$ such that $B(x; r) \subseteq G$. The closed ball $\text{cl } B(x; r/2)$ is a closed subset of $\text{cl } G$. The converse follows directly from (a). $\square$

Lemma 3.31. *Let X be locally compact, let $K \subseteq X$ be compact, and let G be an open set containing K . Then there is an open set U such that*

$$K \subseteq U \subseteq \text{cl } U \subseteq G$$

and $\text{cl } U$ is compact.

Proof. For each $x \in K$, choose an open neighborhood U_x such that $\text{cl } U_x$ is compact and contained in G . Select finitely many $U_{x_1}, \dots, U_{x_n}$ covering K and put $U = U_{x_1} \cup \dots \cup U_{x_n}$. Then

$$\text{cl } U \subseteq \bigcup_{k=1}^n \text{cl } U_{x_k} \subseteq G,$$

and the finite union on the right is compact. $\square$

Theorem 3.32 (Compactly supported bump function). *Let X be locally compact, let $K \subseteq X$ be compact, and let G be open with $K \subseteq G$. There is a continuous function $f : X \rightarrow [0, 1]$ such that*

$$f = 1 \text{ on } K, \quad f = 0 \text{ on } X \setminus G,$$

and f has compact support.

Proof. Choose an open set U such that

$$K \subseteq U \subseteq \text{cl}U \subseteq G$$

and $\text{cl}U$ is compact. The compact Hausdorff space $\text{cl}U$ is normal. By Urysohn's lemma, there is a continuous $g : \text{cl}U \rightarrow [0, 1]$ that equals 1 on K and 0 on $\text{cl}U \setminus U$. Define $f = g$ on $\text{cl}U$ and $f = 0$ on $X \setminus U$. The two definitions agree on their overlap, so the pasting lemma gives continuity. Moreover, $\text{spt } f \subseteq \text{cl}U$. $\square$

Corollary 3.33. *Every locally compact Hausdorff space is completely regular.*

Proof. If F is closed and $x \notin F$, apply the bump-function theorem to $K = \{x\}$ and $G = X \setminus F$. $\square$

Theorem 3.34. *Let X be locally compact, let $K \subseteq X$ be compact, let G be open with $K \subseteq G$, and let $f : K \rightarrow [0, 1]$ be continuous. Then there is a continuous function $F : X \rightarrow [0, 1]$ such that*

$$F|_K = f \quad \text{and} \quad F = 0 \quad \text{on} \quad X \setminus G.$$

The function F may be chosen to have compact support.

Proof. Choose an open set U such that

$$K \subseteq U \subseteq \text{cl}U \subseteq G$$

and $\text{cl}U$ is compact. On the closed subset

$$A = K \cup (\text{cl}U \setminus U)$$

of $\text{cl}U$, define $h = f$ on K and $h = 0$ on $\text{cl}U \setminus U$. The two pieces are disjoint and closed, so h is continuous. Tietze's theorem extends h to a continuous $\tilde{h} : \text{cl}U \rightarrow [0, 1]$. Paste $\tilde{h}$ with the zero function on $X \setminus U$. The resulting function has support contained in $\text{cl}U$. $\square$

Theorem 3.35 (Baire theorem for locally compact spaces). *Let X be locally compact, and let $U_1, U_2, \dots$ be dense open subsets of X . Then*

$$\bigcap_{n=1}^{\infty} U_n$$

is dense in X .

Proof. Let O be a nonempty open subset of X . Put $V_0 = O$. Inductively, the set $V_{n-1} \cap U_n$ is nonempty and open. Choose a nonempty open set V_n such that $\text{cl}V_n$ is compact and

$$\text{cl}V_n \subseteq V_{n-1} \cap U_n.$$

The nonempty compact sets $\text{cl}V_n$ are nested. Their intersection is nonempty, because they are closed subsets of the compact set $\text{cl}V_1$ with the finite-intersection property. Every point of the intersection belongs to $O \cap \bigcap_n U_n$. $\square$

Definition 3.36. Let X be locally compact. A continuous function $\phi : X \rightarrow \mathbb{R}$ vanishes at infinity if, for every $\varepsilon > 0$, the set

$$\{x \in X : |\phi(x)| \geq \varepsilon\}$$

is compact.

The support of a continuous function is

$$\text{spt } \phi = \text{cl}\{x \in X : \phi(x) \neq 0\}.$$

Let $C_0(X)$ be the space of continuous functions that vanish at infinity, and let $C_c(X)$ be the space of continuous functions with compact support.

Theorem 3.37. *Let X be locally compact.*

- (a) *Both $C_0(X)$ and $C_c(X)$ are subalgebras of $C_b(X)$.*
- (b) *$C_0(X)$ is closed in $C_b(X)$ for the supremum metric.*
- (c) *$C_c(X)$ is dense in $C_0(X)$.*

Proof. (a) A function that vanishes at infinity is bounded: it is bounded on the compact set where $|f| \geq 1$ and has absolute value less than 1 elsewhere. For $f, g \in C_0(X)$,

$$\{|f + g| \geq \varepsilon\} \subseteq \{|f| \geq \varepsilon/2\} \cup \{|g| \geq \varepsilon/2\},$$

and

$$\{|fg| \geq \varepsilon\} \subseteq \{|f| \geq \sqrt{\varepsilon}\} \cup \{|g| \geq \sqrt{\varepsilon}\}.$$

Thus sums and products vanish at infinity; scalar multiples are immediate. For compactly supported functions, the support of a sum or product is contained in a finite union of compact supports.

(b) Suppose $f_n \in C_0(X)$ and $f_n \rightarrow f$ uniformly. Given $\varepsilon > 0$, choose N with $\|f - f_N\|_\infty < \varepsilon/2$. Outside the compact set

$$K = \{x : |f_N(x)| \geq \varepsilon/2\},$$

we have $|f(x)| < \varepsilon$. Hence $f \in C_0(X)$.

(c) Let $f \in C_0(X)$ and $\varepsilon > 0$. The set

$$K = \{x : |f(x)| \geq \varepsilon/2\}$$

is compact. Choose a compactly supported function $\psi : X \rightarrow [0, 1]$ with $\psi = 1$ on K . Then $f\psi \in C_c(X)$ and

$$\|f - f\psi\|_\infty \leq \varepsilon/2 < \varepsilon.$$

□

Theorem 3.38. *If (X, d) is a locally compact metric space, then every function in $C_0(X)$ is uniformly continuous.*

Proof. Let $f \in C_0(X)$ and $\varepsilon > 0$. Put

$$K = \{x : |f(x)| \geq \varepsilon/3\}.$$

If K is empty, the conclusion is immediate. Otherwise, choose an open set V with $K \subseteq V$ and compact closure. Since K is compact and $X \setminus V$ is closed and disjoint from K ,

$$\delta_0 = d(K, X \setminus V) > 0$$

when $X \setminus V$ is nonempty; set $\delta_0 = 1$ otherwise. The restriction of f to the compact set $\text{cl } V$ is uniformly continuous, so there is $\delta_1 > 0$ such that

$$x, y \in \text{cl } V, \quad d(x, y) < \delta_1 \implies |f(x) - f(y)| < \varepsilon.$$

Take $\delta = \min(\delta_0, \delta_1)$. If $d(x, y) < \delta$ and at least one of x, y lies in K , then both lie in V , so the compact-set estimate applies. If neither lies in K , then

$$|f(x) - f(y)| \leq |f(x)| + |f(y)| < \frac{2\varepsilon}{3}.$$

Thus f is uniformly continuous. □

Theorem 3.39 (One-point compactification). *Let X be locally compact Hausdorff, and let $\infty \notin X$. Put $X_\infty = X \cup \{\infty\}$. There is a compact Hausdorff topology on X_∞ characterized as follows:*

- (a) X is an open subspace with its original topology;
- (b) a set U containing ∞ is open if and only if $X_\infty \setminus U$ is a compact subset of X ;
- (c) if $\phi \in C_0(X)$ and

$$\tilde{\phi}(x) = \begin{cases} \phi(x), & x \in X, \\ 0, & x = \infty, \end{cases}$$

then $\tilde{\phi} \in C(X_\infty)$. Conversely, restricting a continuous function $F : X_\infty \rightarrow \mathbb{R}$ with $F(\infty) = 0$ gives a member of $C_0(X)$.

Proof. Declare the open subsets of X_∞ to be the open subsets of X together with the sets containing ∞ whose complements are compact in X . Compact subsets of the Hausdorff space X are closed, and a direct check shows that these sets form a topology.

To prove compactness, let $\mathcal{U}$ be an open cover of X_∞ and choose $U_0 \in \mathcal{U}$ containing ∞ . The compact set $X_\infty \setminus U_0$ is covered by finitely many other members of $\mathcal{U}$; together with U_0 they form a finite subcover.

The space is Hausdorff. Two points of X can be separated in X . If $x \in X$, choose an open neighborhood V of x with compact closure. Then

$$V \quad \text{and} \quad X_\infty \setminus \text{cl } V$$

are disjoint open neighborhoods of x and ∞ .

For (c), continuity away from ∞ is clear. For $\varepsilon > 0$, the complement of

$$\{\infty\} \cup \{x \in X : |\phi(x)| < \varepsilon\}$$

is compact, so this is a neighborhood of ∞ mapped into $(-\varepsilon, \varepsilon)$. Conversely, if $F(\infty) = 0$, then

$$\{x \in X : |F(x)| \geq \varepsilon\}$$

is a closed subset of the compact space X_∞ that does not contain ∞ , and is therefore compact in X . □

Remark. If X is noncompact, then X is dense in X_∞ , so this construction is a compactification in the usual sense. If X is compact, the same construction simply adjoins an isolated point ∞ .

Definition 3.40. When X is noncompact, the space X_∞ is called the one-point compactification of X . A space is σ -compact if it is a countable union of compact subsets.

Theorem 3.41. *Let (X, d) be a locally compact metric space. The following are equivalent.*

- (a) X_∞ is metrizable.
- (b) X is σ -compact.
- (c) The metric space $C_0(X)$ is separable.

Proof. Assume (a), and let d_∞ be a compatible metric on X_∞ . Then

$$K_n = \{x \in X : d_\infty(x, \infty) \geq 1/n\}$$

is compact and $X = \bigcup_n K_n$, proving (b).

Assume (b). By repeatedly applying the compact-neighborhood lemma, construct compact sets K_n such that

$$K_n \subseteq \text{int } K_{n+1} \quad \text{and} \quad X = \bigcup_n K_n.$$

A σ -compact metric space is separable and hence second countable. Every compact subset of X is contained in some K_n , because the open sets $\text{int } K_n$ cover X . Hence a countable base for X , together with the neighborhoods $X_\infty \setminus K_n$ of ∞ , is a countable base for X_∞ . Thus X_∞ is compact Hausdorff and second countable, so the Urysohn metrization theorem makes it metrizable. This proves (a).

If (a) holds, the standard separability theorem for $C(K)$ on a compact metric space shows that $C(X_\infty)$ is separable. The closed subspace

$$\{F \in C(X_\infty) : F(\infty) = 0\}$$

is isometric to $C_0(X)$, proving (c).

Finally, assume (c), and let (f_n) be dense in $C_0(X)$. For every $x \in X$, choose a compactly supported $g \in C_0(X)$ with $g(x) = 1$. Some f_n satisfies $\|f_n - g\|_\infty < 1/2$, so $|f_n(x)| > 1/2$. Therefore

$$X = \bigcup_{n=1}^{\infty} \{x \in X : |f_n(x)| \geq 1/2\},$$

and every set on the right is compact. Hence (b) holds. □

Corollary 3.42. *If X is compact Hausdorff, then $C(X)$ is separable if and only if X is metrizable.*

Proof. If X is compact metrizable, the standard separability theorem for $C(K)$ gives separability of $C(X)$.

Conversely, let (f_n) be dense in $C(X)$. This family separates points: if $x \neq y$, choose $f \in C(X)$ with $f(x) \neq f(y)$ and approximate f closely enough by some f_n to preserve the inequality. After composing each f_n with a bounded injective continuous map from $\mathbb{R}$ into $(0, 1)$, the evaluation map

$$e : X \longrightarrow [0, 1]^{\mathbb{N}}, \quad e(x) = (f_1(x), f_2(x), \dots),$$

is continuous and injective. A continuous injection from compact X into the Hausdorff metrizable space $[0, 1]^{\mathbb{N}}$ is a homeomorphism onto its image. Thus X is metrizable. □

3.6 Paracompactness

Definition 3.43. Let X be a topological space. A collection $\mathcal{S}$ of subsets of X is locally finite if every $x \in X$ has a neighborhood that meets only finitely many members of $\mathcal{S}$.

A collection $\mathcal{D}$ refines a collection $\mathcal{S}$ if every member of $\mathcal{D}$ is contained in some member of $\mathcal{S}$.

A Hausdorff space X is paracompact if every open cover of X has a locally finite open refinement that still covers X .

Theorem 3.44. *Every closed subspace of a paracompact space is paracompact.*

Proof. Let F be closed in a paracompact space X , and let $\mathcal{D}$ be an open cover of F . For each $D \in \mathcal{D}$, choose an open set $G_D \subseteq X$ with $D = F \cap G_D$. Then

$$\{G_D : D \in \mathcal{D}\} \cup \{X \setminus F\}$$

is an open cover of X . Let $\mathcal{V}$ be a locally finite open refinement. The nonempty sets $F \cap V$, $V \in \mathcal{V}$, form a locally finite open cover of F . Any such set is contained in some $D \in \mathcal{D}$, because a member of $\mathcal{V}$ contained in $X \setminus F$ has empty intersection with F . $\square$

Theorem 3.45. *Every paracompact Hausdorff space is normal.*

Proof. We first prove regularity. Let F be closed and let $c \notin F$. For each $x \in F$, the Hausdorff property gives an open neighborhood G_x of x such that $c \notin \text{cl } G_x$. Apply paracompactness to the open cover

$$\{G_x : x \in F\} \cup \{X \setminus F\}$$

and let $\mathcal{V}$ be a locally finite open refinement. Let V be the union of the members of $\mathcal{V}$ that meet F . Such a member is contained in some G_x , and local finiteness gives

$$\text{cl } V = \bigcup \{\text{cl } W : W \in \mathcal{V}, W \cap F \neq \emptyset\}.$$

Consequently, $c \notin \text{cl } V$. Thus $F \subseteq V$ and the regularity criterion is satisfied.

Now let A and B be disjoint closed sets. By regularity, for every $a \in A$ choose an open G_a such that

$$a \in G_a \subseteq \text{cl } G_a \subseteq X \setminus B.$$

Apply paracompactness to

$$\{G_a : a \in A\} \cup \{X \setminus A\}.$$

The union U of the members of a locally finite refinement that meet A contains A , and the same closure argument gives $\text{cl } U \cap B = \emptyset$. Hence U and $X \setminus \text{cl } U$ are disjoint open neighborhoods of A and B . $\square$

Lemma 3.46. *Suppose every open cover of a space X has a locally finite closed refinement that covers X . Then X is paracompact.*

Proof. Let $\mathcal{G}$ be an open cover. Choose a locally finite closed cover $\{F_i : i \in I\}$ refining it, with $F_i \subseteq G_i \in \mathcal{G}$. For each $x \in X$, choose an open neighborhood W_x that meets only finitely many F_i . Apply the hypothesis again to the cover $\{W_x : x \in X\}$ and obtain a locally finite closed cover $\mathcal{C}$ refining it. Every $C \in \mathcal{C}$ meets only finitely many F_i .

For $i \in I$, define

$$V_i = X \setminus \bigcup \{C \in \mathcal{C} : C \cap F_i = \emptyset\}.$$

A locally finite union of closed sets is closed, so V_i is open. Also, $F_i \subseteq V_i$, and therefore the V_i cover X .

The family $\{V_i : i \in I\}$ is locally finite. Indeed, choose an open neighborhood O of a point that meets only finitely many members $C_1, \dots, C_m$ of $\mathcal{C}$. If $O \cap V_i \neq \emptyset$, then some C_j meeting O must meet F_i . Each C_j meets only finitely many of the sets F_i , so only finitely many V_i can meet O .

Finally, $\{V_i \cap G_i : i \in I\}$ is a locally finite open refinement of $\mathcal{G}$ and covers X , because $F_i \subseteq V_i \cap G_i$. $\square$

Lemma 3.47 (Michael's refinement lemma). *Let X be regular. Suppose every open cover of X has a locally finite refinement that covers X , without any requirement that the refining sets be open or closed. Then X is paracompact.*

Proof. Let $\mathcal{G}$ be an open cover. For every $x \in X$, choose $G_x \in \mathcal{G}$ and an open set U_x such that

$$x \in U_x \subseteq \text{cl } U_x \subseteq G_x.$$

By hypothesis, the cover $\{U_x : x \in X\}$ has a locally finite refinement $\mathcal{C}$ that covers X . Then

$$\{\text{cl } C : C \in \mathcal{C}\}$$

is a locally finite closed cover. If $C \subseteq U_x$, then $\text{cl } C \subseteq G_x$, so this closed cover refines $\mathcal{G}$. The preceding lemma now proves paracompactness. $\square$

Theorem 3.48 (Michael's theorem). *Let X be regular. Suppose every open cover of X has an open refinement $\mathcal{A}$ that covers X and can be written as*

$$\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{A}_n,$$

where every $\mathcal{A}_n$ is locally finite. Then X is paracompact.

Proof. Fix an open cover and choose such a refinement $\mathcal{A}$. Put

$$U_n = \bigcup \mathcal{A}_n, \quad P_0 = X, \quad P_n = X \setminus \bigcup_{k=1}^n U_k,$$

and

$$D_n = P_{n-1} \setminus P_n = P_{n-1} \cap U_n.$$

The sets D_n are pairwise disjoint and cover X . They also form a locally finite family. Indeed, if $x \notin P_m$, then the open neighborhood $X \setminus P_m$ of x misses every D_n with $n > m$.

Now define

$$\mathcal{D} = \{D_n \cap A : n \geq 1, A \in \mathcal{A}_n\}.$$

This is a cover refining $\mathcal{A}$. It is locally finite: near a fixed point only finitely many D_n occur, and for each of those indices the family $\mathcal{A}_n$ is locally finite. Michael's refinement lemma therefore implies that X is paracompact. $\square$

Corollary 3.49. *Let X be regular. Suppose*

$$X = \bigcup_{n=1}^{\infty} X_n,$$

where every X_n is closed and paracompact and

$$X = \bigcup_{n=1}^{\infty} \text{int } X_n.$$

Then X is paracompact.

Proof. Let $\mathcal{G}$ be an open cover of X . For each n , restrict $\mathcal{G}$ to X_n and choose a locally finite relative-open refinement $\mathcal{B}_n$ covering X_n . If $B \in \mathcal{B}_n$, choose an open set $V_B \subseteq X$ with $B = V_B \cap X_n$, and choose $G_B \in \mathcal{G}$ with $B \subseteq G_B$. The family

$$\mathcal{A}_n = \{V_B \cap \text{int } X_n \cap G_B : B \in \mathcal{B}_n\}$$

is an open family in X , refines $\mathcal{G}$, is locally finite in X , and covers $\text{int } X_n$. Thus $\bigcup_n \mathcal{A}_n$ is a σ -locally finite open refinement covering X . Michael's theorem applies. $\square$

Corollary 3.50. *Every separable metric space is paracompact.*

Proof. A separable metric space has a countable base $\mathcal{B}$. Given an open cover $\mathcal{G}$, let

$$\mathcal{A} = \{B \in \mathcal{B} : B \subseteq G \text{ for some } G \in \mathcal{G}\}.$$

Then $\mathcal{A}$ is a countable open refinement covering X . Write it as a countable union of one-element, hence locally finite, families and apply Michael's theorem. $\square$

Remark. In fact every metric space is paracompact, but proving the full theorem requires a more elaborate argument.

Corollary 3.51. *Every regular σ -compact space is paracompact.*

Proof. A σ -compact space is Lindelöf: from an open cover, choose finitely many sets covering each compact member of a countable compact cover. Thus every open cover has a countable subcover. A countable cover is a countable union of one-element locally finite families, so Michael's theorem applies. $\square$

Theorem 3.52. *Let X be locally compact Hausdorff. Then X is paracompact if and only if it is a disjoint union of open σ -compact subspaces.*

Proof. Suppose first that

$$X = \coprod_{i \in I} X_i,$$

where each X_i is open and σ -compact. Every X_i is locally compact and regular, hence paracompact by the preceding corollary. Given an open cover of X , choose a locally finite refinement on each X_i . Their union is locally finite in X , because each point has the open neighborhood X_i . Thus X is paracompact.

Conversely, suppose X is paracompact. Start with the cover by relatively compact open sets and choose a locally finite open refinement $\mathcal{U}$. Then $\text{cl } U$ is compact for every $U \in \mathcal{U}$. Form a graph whose vertices are the members of $\mathcal{U}$, joining U and V when $U \cap V \neq \emptyset$.

Every vertex has finite degree. Indeed, the compact set $\text{cl } U$ meets only finitely many members of the locally finite family $\mathcal{U}$. Consequently, every connected component of the graph is countable.

For a graph component $\mathcal{C}$, put

$$X_{\mathcal{C}} = \bigcup_{U \in \mathcal{C}} U.$$

These sets are pairwise disjoint, open, and cover X . They are also closed. If $x \in \text{cl } X_{\mathcal{C}}$, local finiteness implies that $x \in \text{cl } U$ for some $U \in \mathcal{C}$. Any member V of the cover containing x meets U , so $V \in \mathcal{C}$ and $x \in X_{\mathcal{C}}$.

Finally, $\text{cl } U \subseteq X_{\mathcal{C}}$ for $U \in \mathcal{C}$ by the same argument. Hence

$$X_{\mathcal{C}} = \bigcup_{U \in \mathcal{C}} \text{cl } U$$

is a countable union of compact sets. □

Definition 3.53. A partition of unity on X is a family of continuous functions $\{\phi_i : X \rightarrow [0, 1] : i \in I\}$ such that

(a) the cozero sets

$$\{x \in X : \phi_i(x) > 0\}, \quad i \in I,$$

form a locally finite family;

(b) for every $x \in X$,

$$\sum_{i \in I} \phi_i(x) = 1.$$

The sum is locally finite and is therefore well defined.

The partition is subordinate to an open cover $\mathcal{G}$ if every cozero set $\{\phi_i > 0\}$ is contained in some member of $\mathcal{G}$.

Lemma 3.54 (Shrinking lemma). *Let X be paracompact and let $\mathcal{U}$ be a locally finite open cover of X . For every $U \in \mathcal{U}$ there is an open set W_U such that*

$$\text{cl } W_U \subseteq U,$$

and the family $\{W_U : U \in \mathcal{U}\}$ is a locally finite open cover of X .

Proof. A paracompact Hausdorff space is normal and hence regular. For every $x \in X$, choose $U(x) \in \mathcal{U}$ containing x and an open set A_x such that

$$x \in A_x \subseteq \text{cl } A_x \subseteq U(x).$$

Choose a locally finite open refinement $\mathcal{V}$ of the cover $\{A_x : x \in X\}$. For every $V \in \mathcal{V}$, choose $x(V)$ such that $V \subseteq A_{x(V)}$ and assign V to $U(x(V))$. Define

$$W_U = \bigcup \{V \in \mathcal{V} : V \text{ is assigned to } U\}.$$

The sets W_U are open and cover X . Since $W_U \subseteq U$ and $\mathcal{U}$ is locally finite, the family (W_U) is locally finite. Also, local finiteness of $\mathcal{V}$ gives

$$\text{cl } W_U = \bigcup \{\text{cl } V : V \text{ is assigned to } U\} \subseteq U.$$

□

Lemma 3.55. *Let X be normal, let $\mathcal{U}$ be a locally finite open cover, and suppose $\{W_U : U \in \mathcal{U}\}$ is an open cover satisfying $\text{cl } W_U \subseteq U$. Then there is a partition of unity subordinate to $\mathcal{U}$.*

Proof. For each $U \in \mathcal{U}$, Urysohn's lemma gives a continuous function $g_U : X \rightarrow [0, 1]$ such that

$$g_U = 1 \text{ on } \text{cl } W_U, \quad g_U = 0 \text{ on } X \setminus U.$$

Because $\mathcal{U}$ is locally finite, the family (g_U) is locally finite, and

$$g = \sum_{U \in \mathcal{U}} g_U$$

is continuous. Since the W_U cover X , we have $g \geq 1$. Define

$$\phi_U = \frac{g_U}{g}.$$

Then $(\phi_U)_{U \in \mathcal{U}}$ is a partition of unity, and $\{\phi_U > 0\} \subseteq U$. □

Theorem 3.56. *A Hausdorff space X is paracompact if and only if every open cover of X admits a subordinate partition of unity.*

Proof. Suppose X is paracompact and let $\mathcal{G}$ be an open cover. Choose a locally finite open refinement $\mathcal{U}$. The shrinking lemma and the preceding normalization lemma produce a partition of unity subordinate to $\mathcal{U}$, hence subordinate to $\mathcal{G}$.

Conversely, if an open cover has a subordinate partition of unity, its cozero sets form a locally finite open refinement covering X . Thus X is paracompact. □

A Zorn's lemma

Definition A.1. A *partial order* on a set S is a relation $\leq$ on S satisfying, for all $x, y, z \in S$:

- (a) $x \leq x$ (reflexivity);
- (b) if $x \leq y$ and $y \leq x$, then $x = y$ (antisymmetry);
- (c) if $x \leq y$ and $y \leq z$, then $x \leq z$ (transitivity).

The pair $(S, \leq)$ is called a *partially ordered set*, or a *poset*.

Let $T \subseteq S$. An element $u \in S$ is an *upper bound* for T if $t \leq u$ for every $t \in T$; the upper bound need not belong to T . An element $m \in T$ is a *maximal element* of T if, whenever $m \leq t$ for some $t \in T$, one has $t = m$.

The poset $(S, \leq)$ is *linearly ordered* (or *totally ordered*) if, for every $x, y \in S$, either $x \leq y$ or $y \leq x$. A *chain* in S is a subset of S that is linearly ordered by the restriction of $\leq$.

Theorem A.2 (Zorn's lemma). *Let $(S, \leq)$ be a nonempty partially ordered set. If every chain in S has an upper bound in S , then S has a maximal element.*

Theorem A.3. *Every vector space over $\mathbb{R}$ has a basis.*

Proof. Let V be a vector space over $\mathbb{R}$, and let

$$\mathcal{P} = \{A \subseteq V : A \text{ is linearly independent}\},$$

ordered by inclusion. The set $\mathcal{P}$ is nonempty because $\emptyset \in \mathcal{P}$.

Let $\mathcal{C}$ be a chain in $\mathcal{P}$, and set

$$B = \bigcup_{A \in \mathcal{C}} A.$$

We claim that B is linearly independent. Suppose $x_1, \dots, x_n \in B$ are distinct and

$$a_1x_1 + \dots + a_nx_n = 0.$$

For each i , choose $A_i \in \mathcal{C}$ such that $x_i \in A_i$. Because $A_1, \dots, A_n$ belong to a chain, one of these finitely many sets contains all the others. Hence all the vectors $x_1, \dots, x_n$ belong to one linearly independent set A_k , and therefore

$$a_1 = \dots = a_n = 0.$$

Thus B is linearly independent, so $B \in \mathcal{P}$. Moreover, every $A \in \mathcal{C}$ satisfies $A \subseteq B$, so B is an upper bound for $\mathcal{C}$ in $\mathcal{P}$. This also covers the empty chain, for which $B = \emptyset$.

By Zorn's lemma, $\mathcal{P}$ has a maximal element M . It remains to show that M spans V . Suppose, to the contrary, that there is a vector $v \in V \setminus \text{span } M$. We claim that $M \cup \{v\}$ is linearly independent. Suppose $m_1, \dots, m_n \in M$ are distinct and

$$av + b_1m_1 + \dots + b_nm_n = 0.$$

If $a \neq 0$, then

$$v = -a^{-1}(b_1m_1 + \dots + b_nm_n) \in \text{span } M,$$

a contradiction. Hence $a = 0$, and the linear independence of M gives $b_1 = \dots = b_n = 0$. Thus $M \cup \{v\}$ is linearly independent and properly contains M , contradicting the maximality of M .

Therefore $\text{span } M = V$, so M is a basis of V . □